

MICROBIOLOGY

CHAPTER 3: BACTERIAL CELL STRUCTURE



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**Reference:
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Chapter 3: Bacterial Cell Structure

We're going to learn about: (/\●▽●)/*:°◇

1. Difference between prokaryotes and eukaryotes
2. size, shape and arrangement of bacterial cells
3. Bacterial cell envelope, nutrients and their uptake
4. Bacterial cytoplasmic structures
5. External structures and motility
6. Endospores

🌸 Difference between prokaryotes and eukaryotes

Prokaryotes differ from eukaryotes in size and simplicity

- Size: Eukaryotes are much larger than prokaryotes
- Simplicity: most prokaryotes lack internal membrane systems (does not have membrane-bounded organelles) and they don't have a nucleus while eukaryotes have nucleus and membrane-bounded organelles

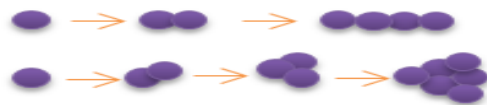
هدول همة الفرقين الاساسين بين ال Eukaryotes (حقيقيات النوى) وال Prokaryotes (بدائيات النوى): الحجم والفرق بالبساطة بتركيبهم وهاد اهم اشئ انو ال eukaryotes عندهم نواة (nucleus) وعضيات محاطة بغشاء اما ال prokaryotes ما عندهم لا نواة ولا عضيات محاطة بغشاء.

🌸 Size, Shape and Arrangement of bacterial cells

To distinguish between different types of bacteria we classify them in terms of size, shape and arrangement:

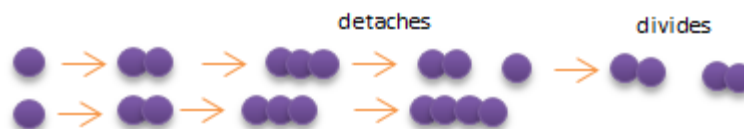
- Shape →
 - a. Cocci (spheres كرات)
 - b. Bacilli (Rods قضبان)
 - c. Various shapes: not just cocci or not just rods

- Arrangement → determined by plane of division
(يعني ممكن تنقسم البكتيريا بمستوى واحد وتعمل ترتيب على شكل خيط او ممكن تنقسم بعدة مستويات وتعطي مثلا شكل زي قطف العنب)



→ Determined by separation or not

(يعني اذا بعد ما يصير الانقسام.. الخلية الجديدة بتنفصل عن الاولى او لا)

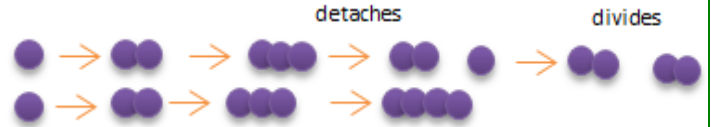


- Size: varies

- Shape and arrangement

- a. Cocci shape and arrangement

- diplococci (s., diplococcus) - pairs
→ أزواج
- streptococci - chains سلاسل →
- staphylococci - grape-like clusters تجمعات مثل قطف العنب
- tetrads - 4 cocci in a square بتجمّعوا أربعة مع بعض على شكل مربع
- sarcinae - cubic configuration of 8 cocci بتجمّعوا 8 مع بعض على شكل مكعب



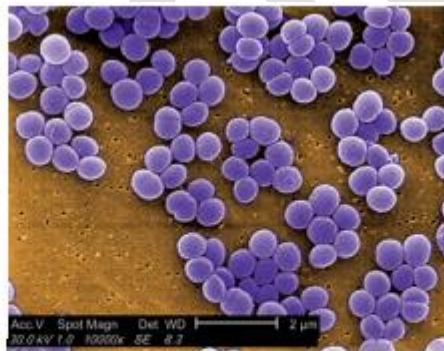
All cocci shaped bacteria are non-motile

- b. Bacilli

- differ considerably in their length-to-width ratio, the **coccobacilli** being so short and wide that they resemble cocci.
- many rods occur singly, some remain together after division to form pairs or chains
some bacilli-shaped bacteria are motile and some are not



(a) *S. agalactiae*—cocci in chains



(b) *S. aureus*—cocci in clusters



(c) *B. megaterium*—rods in chains

Figure 3.1 Cocci and Rods Are the Most Common Bacterial Shapes. (a) *Streptococcus agalactiae*, the cause of Group B streptococcal infections; color-enhanced scanning electron micrograph ($\times 4,800$). (b) *Staphylococcus aureus*; color-enhanced scanning electron micrograph. (c) *Bacillus megaterium*, Gram stain ($\times 1,000$).

- c. Other shapes and arrangements

- **Vibrios** are comma-shaped (figure 3.2a)
→ ex. *Vibrio cholerae* that causes Cholera
→ Its motility is unique; it is called "Shooting star motility" since it's so fast, it rotates 1100 times/sec!
- **Spirilla** are rigid, spiral-shaped cells (figure 3.2b) → **(Rigid helices)**
- **Spirochetes** are flexible, spiral-shaped bacteria that have a unique, internal flagellar arrangement (figure 3.2c) → **(Flexible helices)**
→ one of the largest bacteria → its length may reach $100\mu\text{m}$
→ it is motile but does not have a flagella (رح نشرح عنو بهاد التشابتر). It has a structure embedded in its membrane that rotates all around the bacteria called **Axial fibril (axial**

filament). It moves like a snake so it must be on a hard surface (this movement is called **crawling** زحف)

- **Mycelium** - network of long, multinucleate filaments or threads (figure 3.2e)
- **pleomorphic**, being variable in shape and lacking a single, characteristic form

يعني مثلا ممكن يكونوا cocci بمرحلة معينة من حياتهم بعدين بتغيروا



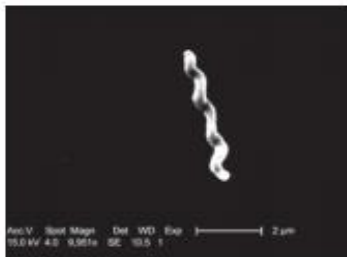
(a) *V. vulnificus* — comma-shaped vibrios



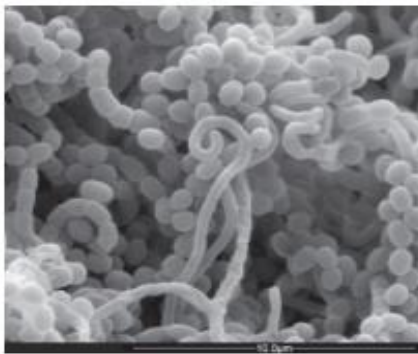
(d) *C. Crescentus*—a stalked bacterium



(f) *C. crocatus* fruiting body



(b) *C. jejuni*—Spiral shaped



(e) *Streptomyces*—a filamentous bacterium



(c) *Leptospira interrogans*—a spiral bacterium

Figure 3.2 Other Cell Shapes and Aggregations. (a) *Vibrio vulnificus*, scanning electron micrograph (SEM, X13,184). (b) *Campylobacter jejuni*, SEM. (c) *Leptospira interrogans*, the spirochete that causes the waterborne disease leptospirosis. (d) *Clostridium crescentus*, SEM. (e) *Streptomyces* sp., SEM. (f) Fruiting body of the myxobacterium *Chondromyces crocatus*. The fruiting body is composed of thousands of cells.

• Size

- The size of the largest bacteria is the same as the size of a human red blood cell
 - ➔ Diameter: 7000 nm (7 μm)
 - ➔ Ex. *Oscillatoria*
 - ➔ Usually large bacteria are photosynthetic bacteria because they need large surface area for light absorption
- The size of the smallest bacteria is the same size as the largest virus → Pox virus
 - ➔ Ex. *Rickettsia* (mycoplasma)
 - ➔ They have a "predatory mechanism" by which they get inside other cells and hide there in order to live (Parasitic) (protective mechanism from predation)

❖ One exception is *Epulopiscium fishelsoni*, which is a very very large bacteria → near the size of a Paramecium (which is a single celled eukaryotic organism)

❖ Usually the size is important for nutrient uptake



Figure 3.4 A Giant Bacterium. This photograph shows *Epulopiscium fishelsoni* dwarfing the paramecia, which are protozoa. *E. fishelsoni* cells are about 0.53 mm long. Phase-contrast microscopy.

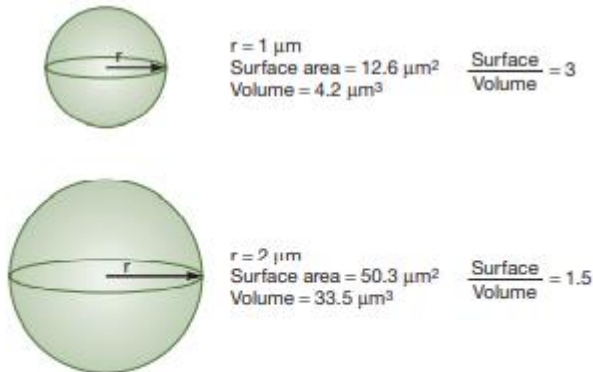


Figure 3.5 The Surface-to-Volume Ratio Is an Important Determinant of Cell Size. Surface area is calculated by the formula $4\pi r^2$. Volume is calculated by the formula $\frac{4}{3}\pi r^3$. Shape also affects the S/V ratio; rods with the same diameter as a coccus have a greater S/V ratio.

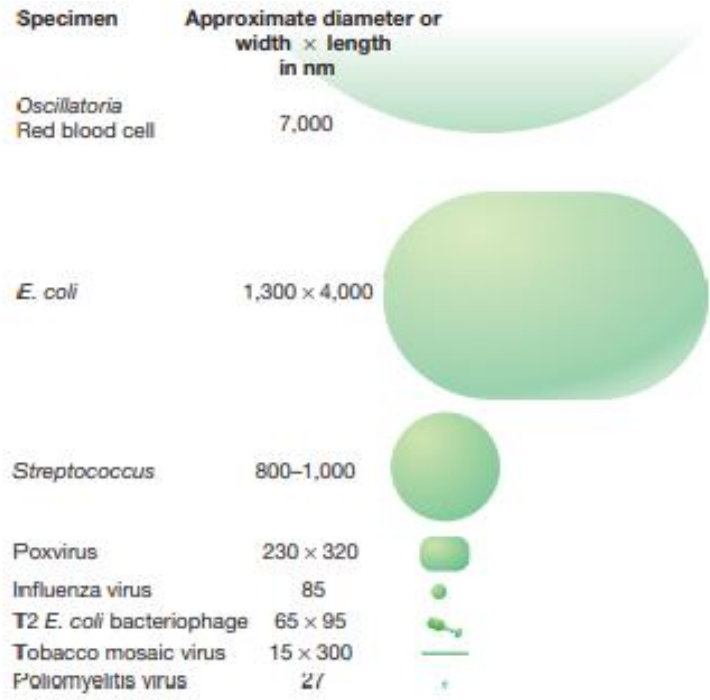


Figure 3.3 Sizes of Bacteria Relative to a Red Blood Cell and Viruses. Recall that 1,000 nm = 1 μm. Thus *E. coli* is $1.3 \times 4 \mu\text{m}$.

➔ Surface to volume ratio (S/V)

خأصنا من حجم ، شكل وترتيب الخلايا البكتيرية وهأا رح نبأش يتركيبها < () >

Bacterial Cell Organization - Common Features:

3 main parts we're going to talk about :

- Cell envelope- 3 layers
- Cytoplasm
- External structures

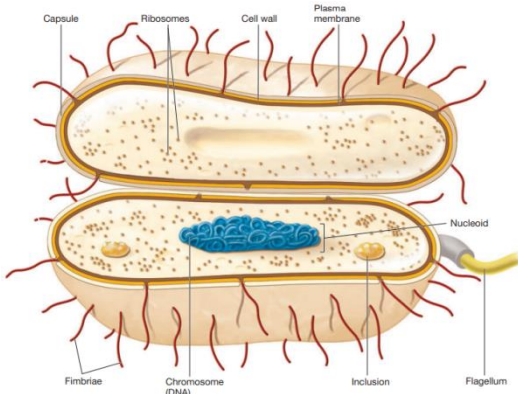


Table 3.1	Common Bacterial Structures and Their Functions
Plasma membrane	Selectively permeable barrier, mechanical boundary of cell, nutrient and waste transport, location of many metabolic processes (respiration, photosynthesis), detection of environmental cues for chemotaxis
Gas vacuole	An inclusion that provides buoyancy for floating in aquatic environments
Ribosomes	Protein synthesis
Inclusions	Storage of carbon, phosphate, and other substances; site of chemical reactions (microcompartments); movement
Nucleoid	Localization of genetic material (DNA)
Periplasmic space	In typical Gram-negative bacteria, contains hydrolytic enzymes and binding proteins for nutrient processing and uptake; in typical Gram-positive bacteria, may be smaller or absent
Cell wall	Protection from osmotic stress, helps maintain cell shape
Capsules and slime layers	Resistance to phagocytosis, adherence to surfaces
Fimbriae and pili	Attachment to surfaces, bacterial conjugation and transformation, twitching
Flagella	Swimming and swarming motility
Endospore	Survival under harsh environmental conditions

❁ Bacterial cell envelope, nutrients and their uptake

📍 A. Cell envelope

Bacterial plasma membranes control what enters and leaves the cell

Bacterial cell envelope consists of:

- 1- Plasma membrane
- 2- Cell wall
- 3- Layers outside the cell wall

1. Plasma membrane

- For any cell to be called a CELL, it has to have a plasma membrane → absolute REQUIRMENT for all living organisms

ما بربط ولا بأي حال من الاحوال يكون عندك خلية وتكون بلا plasma membrane لانو اذا تمت إزالته ، بتطلع محتويات الخلية للبيئة وتبتطل الخلية موجودة بالاضافة الى انو بصير فيه عمليات مهمة الخلية ما بتقدر تعيش بدونها (ـهـــ)

- Encompasses the cytoplasm بلف حول او يطوق السيتوبلازم
- Selectively permeable barrier

ما بسمح لأي اشي يفوت جوا الخلية لهيك منحيلو selective يعني يختار شو يدخل وشو اللي ما يدخل من ايونات ومركبات مختلفة وشو اللي لازم يطلع برا من هدول على شكل waste

- Interacts with external environment
 - receptors for detection of and response to chemicals in surroundings
 - transport systems
 - metabolic processes

بصير فيه عمليات ايض مثل التنفس الخلوي (respiration)، البناء الضوئي (photosynthesis) وتصنيع ال lipids ومكونات الخلية

• Fluid Mosaic Model of Membrane Structure:

- The most widely accepted model for membrane structure is the **fluid mosaic model** which proposes that membranes are Lipid bilayers with floating proteins

- amphipathic lipids:

- polar ends (hydrophilic: interact with water)
- non-polar tails (hydrophobic: insoluble in water)

Amphipathic تعني احتواء المركب على جزئين مختلفين واحد hydrophilic والتاني hydrophobic

- membrane proteins → two types

1. peripheral
 - loosely connected to membrane
 - easily removed
2. integral
 - amphipathic - embedded within membrane
 - carry out important functions

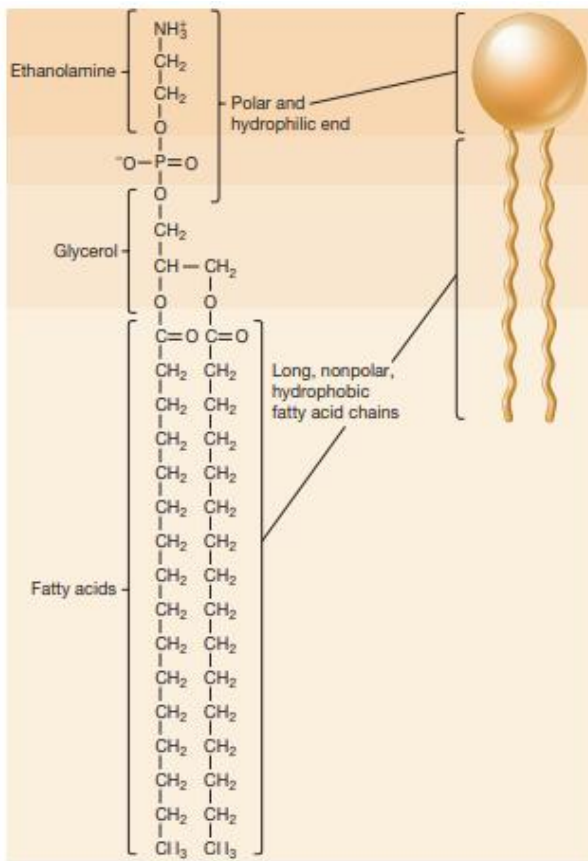


Figure 3.8 The Structure of a Phospholipid.

Phosphatidylethanolamine, a phospholipid often found in bacterial membranes.

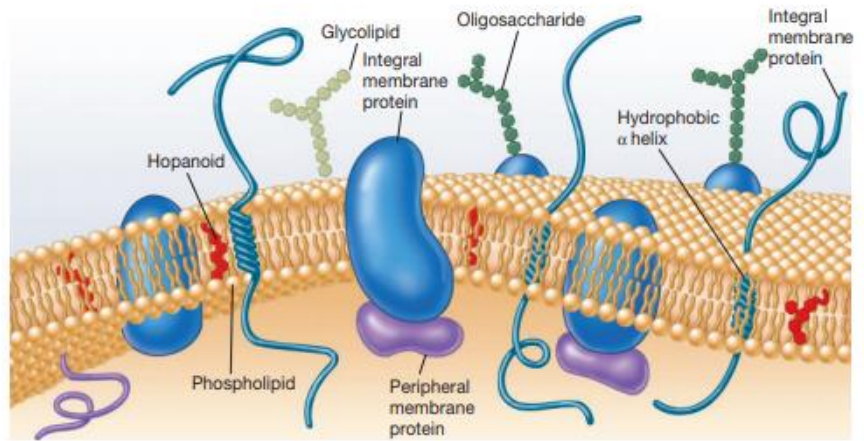


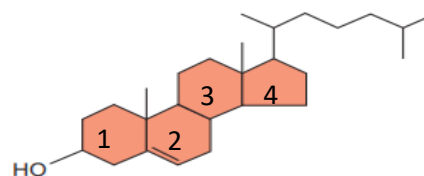
Figure 3.7 The Fluid Mosaic Model of Bacterial Membrane Structure. This diagram shows the integral membrane proteins (blue) floating in a lipid bilayer. Peripheral membrane proteins (purple) are associated loosely with the inner membrane surface. Small spheres represent the hydrophilic ends of membrane phospholipids, and wiggly tails are the hydrophobic fatty acid chains. Other membrane lipids such as hopanoids (red) may be present. Phospholipids are drawn much larger than their actual size.

- Bacterial lipids:

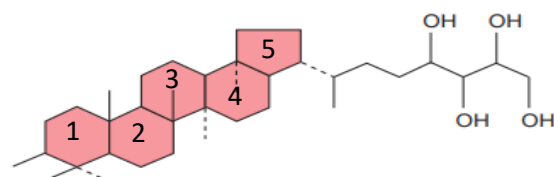
- Saturation levels of membrane lipids reflect environmental conditions such as temperature

للتوضيح: يختلف تكوين الـ lipids مع درجة الحرارة تبعاً للبيئة بطريقة بتخللي الغشاء يضل سائل أثناء نموف عشان يضل الغشاء سائل البكتيريا اللي بتنمو في درجات حرارة منخفضة عندها **unsaturated fatty acids** اكثر بالـ phospholipids اللي بالأغشية يعني يكون في **double bonds** اكثر بالسلاسل الكربونية الطويلة تبعها. في درجات الحرارة العالية ، بتحتوي الـ phospholipids تبعها على **fatty acids** مشبعة اكثر (يعني يكون فيها **single bonds**) وهيك يتم الحفاظ على سيولة الغشاء في درجات الحرارة المختلفة.

- Bacterial membranes lack sterols but do contain sterol-like molecules called **hopanoids** (they don't have Cholesterol!)
- stabilize membrane
- found in petroleum



(a) Cholesterol (a steroid) is found in the membranes of eukaryotes.



(b) Bacteriohopanetetrol (a hopanoid) is found in many bacterial membranes.

Figure 3.9 Membrane Steroids and Hopanoids.

- Uptake of Nutrients -Through the Barrier
 - Bacterial Nutrition:
 - **Macroelements (macronutrients)** → required in relatively large amounts because they are crucial for making important molecules like amino acids and proteins and they are:
 - CHNOPS (C,H,N,O,P,S) (اقرأوها "كنوبس" للحفظ)
 - found in organic molecules (proteins, lipids, carbohydrates, and nucleic acids)
 - K, Ca, Mg, and Fe
 - cations; serve in variety of roles including assisting enzymes and biosynthesis
 - **Micronutrients (trace elements)**
 - required in small amounts
 - often supplied in water/media components.
 - ubiquitous (meaning that they are found everywhere)
 - Work to assist enzymes (cofactors)
 - Mn, Zn, Co, Mo, Ni, and Cu
 - Some unique substances may be required
 - Growth factors:
 - organic compounds
 - essential cell components (or their precursors) that the cell cannot synthesize
 - must be supplied by environment if cell is to survive and reproduce
 - classes include:
 - amino acids→ needed for protein synthesis
 - purines and pyrimidines→needed for nucleic acid synthesis
 - vitamins→function as enzyme cofactors
 - heme
- Methods for Uptake of Nutrients
 - Microbes can only take in dissolved particles across a selectively permeable membrane

انتبهوا انو بس dissolved particles يعني على شكل ذائب ومذيب

1. **Passive diffusion** :Some nutrients enter by passive diffusion→the process by which molecules move from a region of higher concentration to one of lower concentration; that is, the molecules move down the concentration gradient like some gases, including

O_2 and CO_2 , easily cross the plasma membrane by passive diffusion. H_2O also moves across membranes by passive diffusion and NO ENERGY IS NEEDED.

2. Facilitated Diffusion:

Similar to passive diffusion in:

- movement of molecules is not energy dependent
- direction of movement is from high concentration to low concentration

size of concentration gradient impacts rate of uptake

Facilitated vs. Passive Diffusion

Differs from passive diffusion in which

- it uses membrane bound carrier molecules (**permeases**)

نقل المواد بالpassive بصير عبر الغشاء نفسه أما بالfacilitated بنقلوا المواد من خلال transport proteins

منسميها permeases (ومن اسمو Facilitated يعني مسهل باستخدام البروتينات)

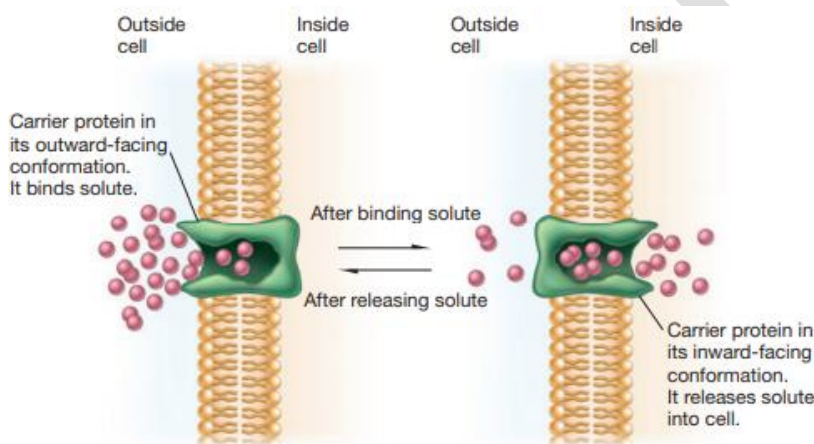


Figure 3.11 A Model of Facilitated Diffusion. Because there is no energy input, molecules continue to enter only as long as their concentration is greater on the outside.

- smaller concentration gradient is required for significant uptake of molecules
- effectively transports glycerol, sugars, and amino acids

بما انو البروتين بوصل لمرحلة اشباع بووقف اخذ المواد لهيك ملاحظ انحنا المنحنى وثباته لفترة من الزمن

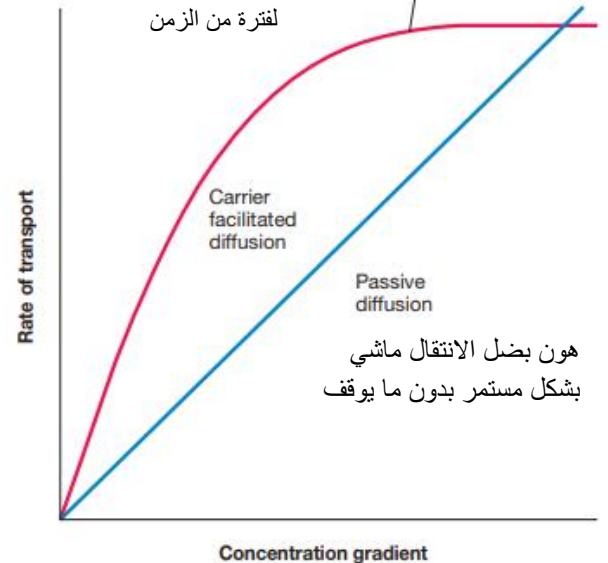


Figure 3.10 Passive and Facilitated Diffusion. The rate of diffusion depends on the size of the solute's concentration gradient (the ratio of the extracellular concentration to the intracellular concentration). This example of facilitated diffusion involves a carrier protein that can be saturated. Sometimes facilitated diffusion is mediated by a channel. Channels often do not exhibit a saturation effect.

أثناء الfacilitated diffusion، يتحرك المواد عبر الغشاء بمساعدة

الtransport proteins التي بتكون إما Channels أو Carriers. بزداد معدل الfacilitated diffusion مع

الconcentration gradient بشكل أسرع بكثير وبتراكيزات أقل من الجزيء الي عم بنقل من الpassive diffusion (الشكل 3.10). ف بضل الcarrier ياخذ كمان من الsolute حتى يوصل لمرحلة اشباع والتي مبيينة بالخط الاحمر بفترة ثبوت بالمنحنى لانو بعد ما يصير مشبع ببطل ياخذ اي اشي زيادة ويعددها بفرغ محتوياته جوا الخلية.

- Facilitated diffusion is more prominent in eukaryotic cells than in bacteria or archaea → why? many bacteria live in environments where nutrient concentrations are

low, and facilitated diffusion cannot concentrate nutrients inside cells. Therefore energy dependent transport mechanisms that do concentrate nutrients are significantly more important uptake mechanisms for bacterial cells.

الفكرة بالنقطة هاي انو نركز على فكرة احتياج ال facilitated diffusion لتركيز المواد جوا البروتين. ف اذا اصلا البيئة الموجودة فيها البكتيريا ما في غذاء كتير ف هاد يعني انو ما رح تقدر تركّز المواد وهاد يعني انو ما رح تستخدم ال facilitated diffusion كتير لهيك منلاقي هاي العملية اكثر بال eukaryotes ورح تعتمد على الطرق الي يتم فيها استخدام الطاقة اكثر وهلا رح نشرح عنهم.

3. Active Transport

- Found in eukaryotes and prokaryotes
- Energy-dependent process either by using energy directly → ATP or indirectly → proton motive force*
- Move molecules against the gradient
يعني من الاقل تركيز للاعلى تركيز
- Concentrates molecules inside cell
لانو زي ما حكينا من قبل، البكتيريا بتعيش ببينة فقيرة بالغذاء لهيك لازم تخزن المواد للاستخدام المستقبلي
- Involves carrier proteins (permeases)
→ carrier saturation effect is observed at high solute concentrations
يعني لما تكتمل حملتهم بوصلوا للإشباع وما بياخدوا كمان غذاء لما يكون تركيزه خارج الغشاء عالي
- **Two types of active transport:**
 - a. Primary active transport
 - Source of energy: They use energy provided by ATP hydrolysis to move substances against a concentration gradient without modifying them
 - Primary active transporters are uniporters; they move a single molecule across the membrane
 - ATP-binding cassette transporters (**ABC transporters**) are important primary active transporters
(ATP binding cassette means that there are structures which bind to ATP to activate the transportation of materials)
 - Observed in Bacteria, Archaea, and eukaryotes
 - Consist of:
 - 2 hydrophobic membrane spanning domains ()
 - 2 cytoplasmic associated ATP-binding domains (hydrophilic)
 - substrate binding domains

**Proton motive force:*

The movement of H^+ in and out the membrane creates an electrical

- TWO ATP molecules are needed in order to go inside the cell (to drive the movement of the solute)

خطوات النقل:

1. يرتبط الـ solute بالـ solute-binding protein

ويمشوا لعند ABC transporter

2. يرتبطوا بالـ transporter وبنفك الارتباط بين

الـ solute والـ solute binding protein (زي انت

لما توصل المكان الي بدك ياه بتنزل من السيارة وبتتركها)

وبتصير عملية الـ hydrolysis لجزيئين من الـ ATP عشان

ينتقل الـ solute عبر الـ transporter.

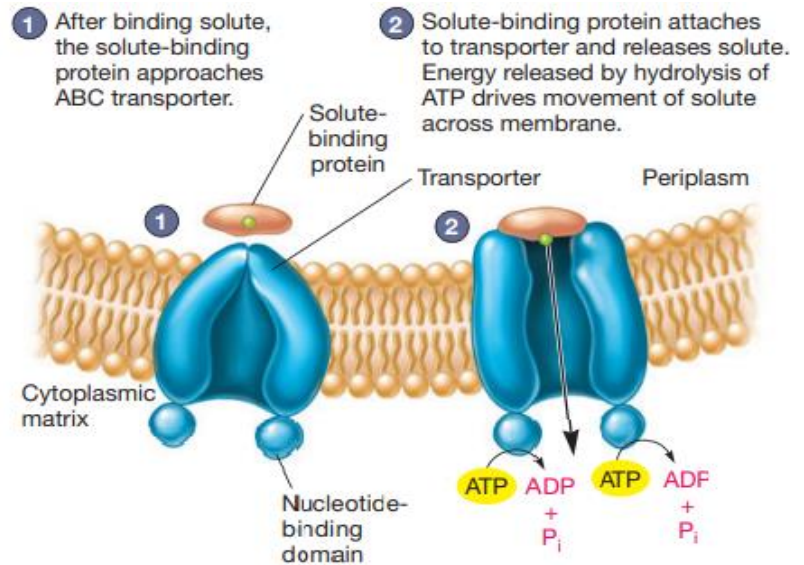


Figure 3.13 ABC Transporter Function. Shown here is a transporter that works with a substrate-binding protein free in the periplasm. Other substrate-binding proteins are associated with the plasma membrane, always associated with the transporter, or even fused to the transporter.

b. Secondary active transport

- Major facilitator superfamily (MFS)*
- Source of energy: couples the potential energy of ion gradients to transport of substances without modifying them.

يعني بتستخدم الـ potential energy الناتجة من الفرق بتركيز الايونات لنقل المواد

**The major facilitator superfamily is a superfamily of membrane transport proteins that facilitate movement of small solutes across cell membranes in response to chemiosmotic gradients.*

- Secondary active transporters are cotransporters

يعني بنقلوا مادتين بنفس الوقت (H^+ الي هو الايون ومادة ثانية)

They can either be:

- Symporters: When the ion and other substance both move in the same direction
- Antiporters: When they move in opposite directions

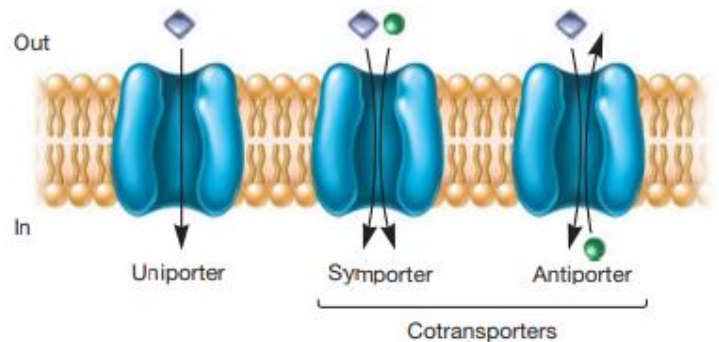


Figure 3.12 Carrier Proteins Can Be Uniporters or Cotransporters.

Uniporters move a single substance into the cell. Cotransporters simultaneously move two substances across the membrane. When both substances move in the same direction, the carrier is a symporter. When the two substances move in opposite directions, the carrier is an antiporter.

يعني لما يتحرك الـ H^+ بنقل معو solute معين. اما بتكون حركتهم مع بعض وبنفس الاتجاه او عكس بعض

4. Group translocation

- Only found in bacteria
- Energy dependent transport that chemically modifies molecule as it is brought into cell
- Best known translocation system is Phosphoenolpyruvate: sugar phosphotransferase system (PTS)

How it works:

- The cell won't uptake
Manitol & Glucose until they
modify their shape.

هدول المركبين الخلية ما بتقدر تاخدھم بشكلھم الاصلي
ولا لازم يتغيروا عشان يقدروا يفوتوا

- How they are modified?

By binding to **phosphate**
(they have to be
phosphorylated)

التغير اللي لازم يصير هو ارتباطھم بمجموعة

phosphate

BUT..from where do they
get it?

From the Inside of the cell

ال phosphate الي بدھم يرتبطوا فيه بيحي من جوا

الخلية وعشان يوصل لازم يرتبط بعدة انزيمات كل واحد فيھم بعطي ال phosphate للي قدامه قبل ما يوصل لل manitol
وال glucose وبناء على هاد الحكي عندي 3 اجزاء مهمين بهاي العملية:

1.Enzyme I

2.Heat stable protein

3.Enzyme II and it has three parts: A, B & C

Phosphoenolpyruvate اسمو مركب اشبي بمركب اسمو

Phosphoenolpyruvate $\xrightarrow{\text{Hands P to}}$ Enzyme 1 $\xrightarrow{\text{Hands P to}}$ HPr (heat-stable protein) $\xrightarrow{\text{Hands P to}}$ Enzyme II A $\rightarrow$
Enzyme II B $\xrightarrow{\text{Hands P to}}$ Enzyme II C $\rightarrow$ Manitol & Glucose

- ❖ The only difference between glucose transmission and manitol transmission is :
Enzyme II A&B are attached in the case of manitol but separated when
transferring Glucose

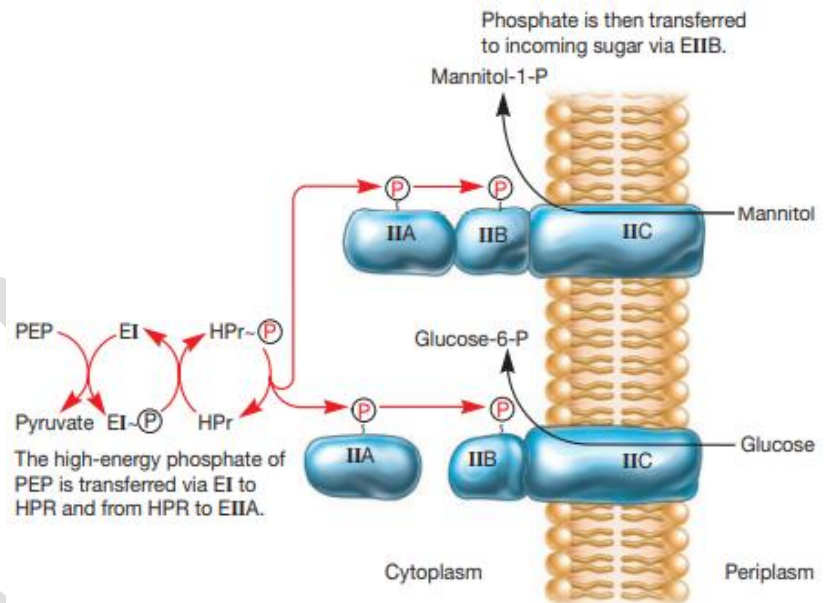


Figure 3.14 Group Translocation: Bacterial PTS Transport. Two examples of the phosphoenolpyruvate: sugar phosphotransferase system (PTS) are illustrated. The following components are involved in the system: phosphoenolpyruvate (PEP), enzyme I (EI), the low molecular weight heat-stable protein (HPr), and enzyme II (EII). EIIA is attached to EIIB in the mannitol transport system and is separate from EIIB in the glucose system.

5. Iron uptake

- Microorganisms require iron for their growth
 - Bacteria can't produce iron so they have to take it from the environment.
- ferric iron is very insoluble so its uptake is difficult (remember: all nutrients should be dissolved) and based on that it can't pass the membrane freely unless it is complexed with **Siderophore**. then, the complex is transported into the cell.

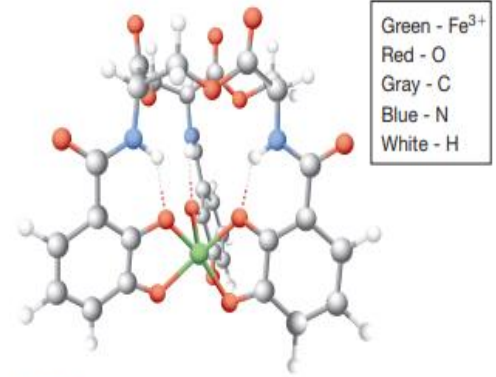


Figure 3.15 Enterobactin: A Siderophore Produced by *E. coli*. Ball-and-stick model of enterobactin complexed with Fe^{3+} .

2. Cell wall

- Absolute requirement for bacterial cells
- Two groups of bacteria → **Gram positive & Gram negative**
- Cell wall functions:
 - 1- maintains shape of the bacterium(almost all bacteria have one)
 - 2- helps protect the cell from osmotic lysis and toxic materials
 - 3- may contribute to pathogenicity

الـ cell wall هو خط الدفاع الاول للبكتيريا . يحافظ على شكلها وبحميها من انها تتفجر نتيجة للضغط الاسموزي وبحميها من المواد السامة بالاضافة الى انو ممكن يكون عامل مهم في قدرتها في التسبب بالامراض .

- The main structure of the cell wall: **Peptidoglycan** (murein)
 - rigid structure lying just outside the cell plasma membrane
 - two types based on Gram stain
 - Gram-positive: stain purple; thick peptidoglycan
 - Gram-negative: stain pink or red; thin peptidoglycan and outer membrane

في طريقة لصبغ البكتيريا اسمها Gram staining لما نصبغ البكتيريا فيها يكون عنا نتيجتين محتملتين بناء على بناء الـ peptidoglycan :

- Gram positive اذا كان الـ Peptidoglycan سميك وهيك بطلع لون البكتيريا بنفسجي
- Gram negative اذا كان الـ peptidoglycan رفيع ويكون في كمان طبقة منسميها outer membrane وهيك بطلع لون البكتيريا من زهري لاحمر

- Peptidoglycan structure

- Meshlike polymer of identical subunits forming long strands
- البناء تبعو مثل الشبكة. هاي الشبكة مكونة من وحدات معينة متجمعة مع بعض على شكل سلاسل (مجموعة السلاسل بتعطي الشبكة الكبيرة)

→ Two alternating sugars

- N-acetylglucosamine (NAG)
- N- acetylmuramic acid (NAM)



→ They bind to four alternating D

- and L- amino acids forming a helix structure

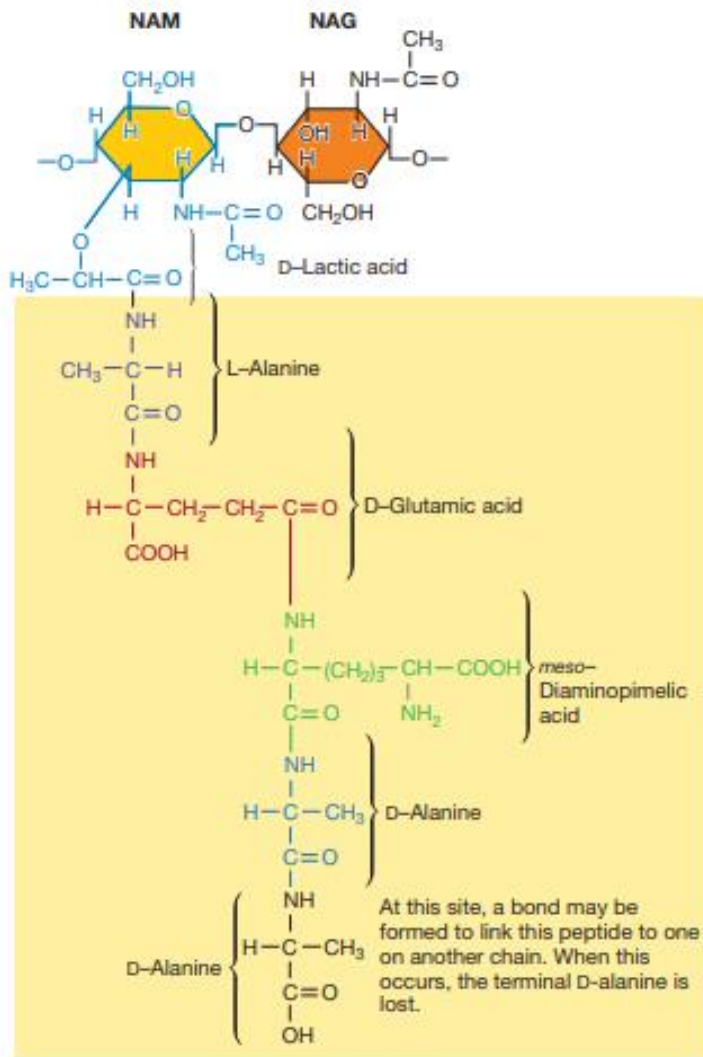


Figure 3.17 Peptidoglycan Subunit Composition. Shown is the peptidoglycan subunit of *E. coli*, many other typical Gram-negative bacteria, and many typical Gram-positive bacteria. This illustration shows the subunit before it has been inserted into the existing peptidoglycan polymer. NAG is *N*-acetylglucosamine. NAM is *N*-acetylmuramic acid. The stem peptide is composed of alternating D- and L-amino acids; it terminates with two D-alanines. The amino acids are shown in different shades of color for clarity.

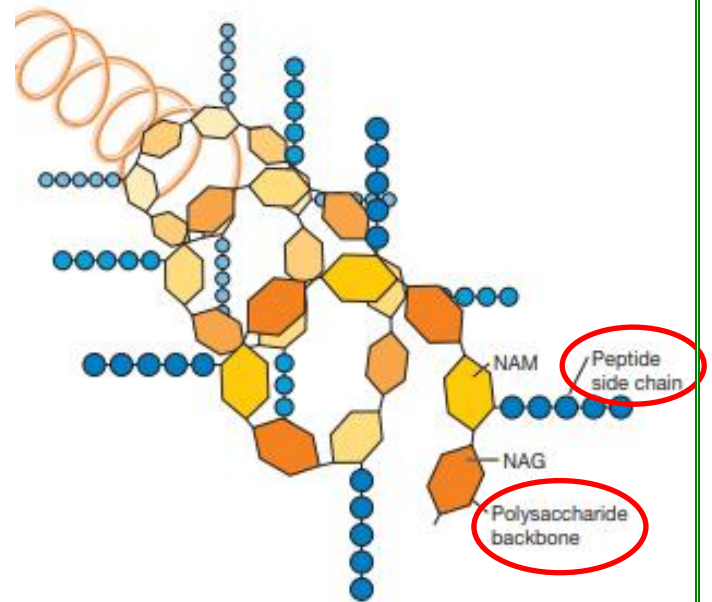


Figure 3.18 A Helical Peptidoglycan Strand. Because of the strand's helical nature, the stem peptides project out in different directions from the M-NAG backbone. Here the stem peptides are shown projecting out 90° angles. Some studies suggest that the angle is actually 120°.

كل NAM و NAG مرتبطين بخمس L-&D-amino acids متبادلين

لما يرتبط هاد ال peptide بسلسلة ثانية .. اخر D-amino acid بتتفك وبصيرو اربعة.

- Peptidoglycan strands have a helical shape
- Peptidoglycan chains are cross-linked by peptides for strength
 - interbridges may form
 - peptidoglycan sacs -interconnected networks
 - various structures occur

(a) Direct linkage → Gram -ve → E. coli

(b) Peptide Interbridge → Gram +ve → *Staphylococcus aureus*

- If Ala binds to Ala instead of Lys, this would give a whole different type of bacteria

ملاحظ بb انو الAla مرتبط بLys. اذا كان مرتبط بأي واحد ثاني غير الLys فهاد رح يعطيني نوع جديد ومختلف كليا عن البكتيريا الاصلية (*~*)

→ Gram Positive cell walls

- Composed primarily of ONE thick layer of peptidoglycan
- The Periplasmic space is small (شرحه بصفحة 16)
- Gram +ve bacteria secrete exo-enzymes to able to degrade nutrients outside the cell and uptake them
- May also contain **teichoic acids** (negatively charged)
 - help maintain cell envelope
 - protect from environmental Substances (detergents or antibiotics)
 - may bind to host cells

→ Gram Negative cell walls

- More complex than Gram-positive
- Consist of two layers of cell wall: a thin layer of Peptidoglycan surrounded by an outer membrane
- Outer membrane : (OM) outside thin peptidoglycan layer composed of lipids, lipoproteins, and lipopolysaccharide (LPS)

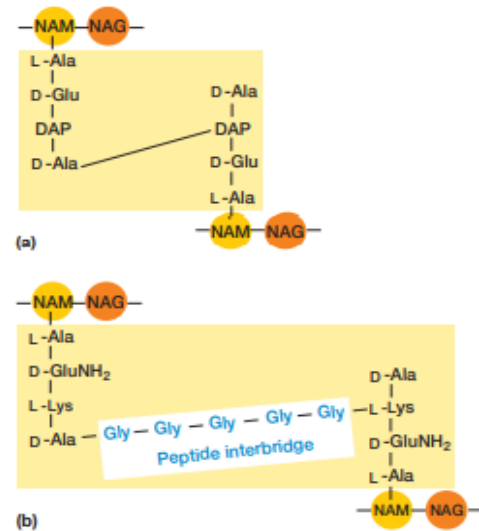
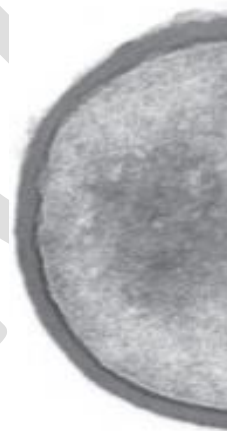
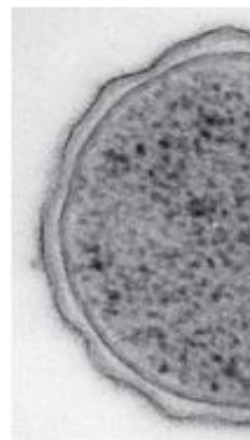
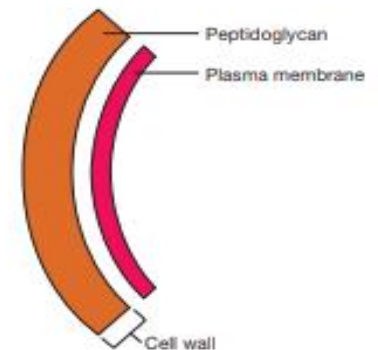


Figure 3.19 Peptidoglycan Cross-Links Can Be Direct or Indirect via a Peptide Interbridge. (a) *E. coli* peptidoglycan with direct cross-linking, typical of many Gram-negative bacteria. (b) *Staphylococcus aureus* peptidoglycan with an interbridge. *S. aureus* stains Gram positive. NAM is N-acetylmuramic acid. NAG is N-acetylglucosamine. Gly is glycine. D-GluNH₂ is D-glutamic acid with an NH₂ group attached to the α carbon (the carbon next to the carboxyl group).



The typical Gram-positive cell envelope



The typical Gram-negative cell envelope

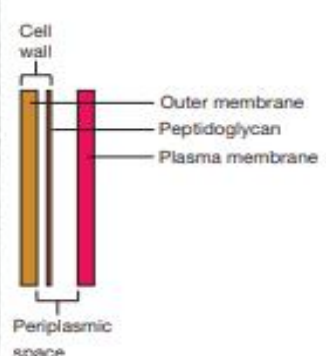


Figure 3.16 Cell Envelopes of Typical Gram-Positive and Gram-Negative Bacteria. Cell envelopes consist of the plasma membrane and any layers (e.g., cell wall) exterior to it. For simplicity, we show only the plasma membrane and cell wall. Slime layers, capsules, and S-layers are common additional layers that lie outside the cell wall. *Staphylococcus aureus* (top) has a typical Gram-positive cell wall that consists primarily of peptidoglycan (also called murein). *Myxococcus xanthus* (bottom) has a typical Gram-negative cell wall consisting of a thin layer of peptidoglycan, an outer membrane, and the periplasmic space.

- No teichoic acids
- Braun's lipoproteins connect OM to peptidoglycan
- Other adhesion sites reported
- Peptidoglycan is ~5-10% of cell wall weight

☞ Periplasmic Space of Gram +ve Bacteria

- Between plasma membrane and cell wall
- Periplasm has relatively few proteins

كتلخيص للموضوع: ال periplasmic space موجود بين ال cell wall وال plasma membrane، مساحته اقل من الي موجود بال gram -ve والبروتينات الي فيه عددها قليل

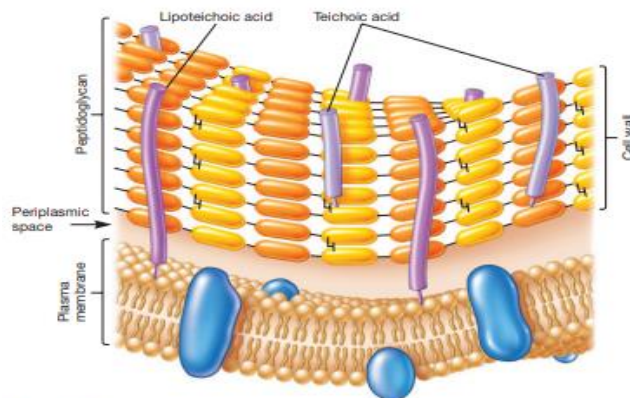


Figure 3.22 Typical Gram-Positive Cell Wall. This component of the cell envelope lies just outside the plasma membrane.

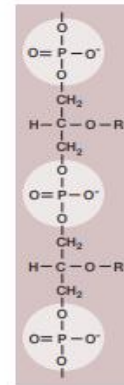


Figure 3.23 Teichoic Acid Structure. The segment of a teichoic acid made of phosphate, glycerol, and a side chain, R. R may represent D-alanine, glucose, or other molecules.

☞ Periplasmic space of Gram -ve bacteria

Periplasmic space differs from that in Gram-positive cells

- may constitute 20-40% of cell volume
- many enzymes present in periplasm
- hydrolytic enzymes, transport proteins and other proteins

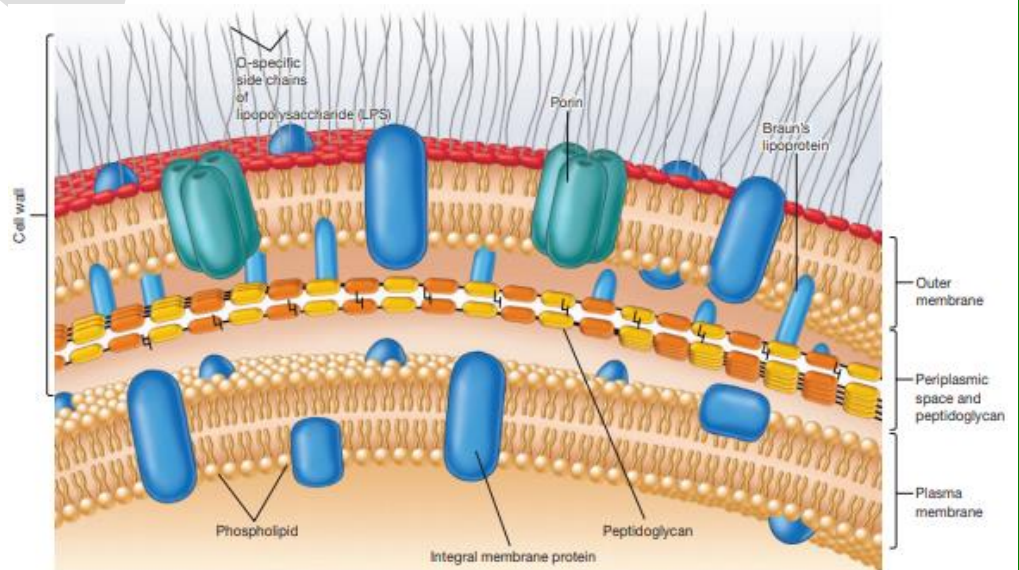


Figure 3.24 Typical Gram-Negative Cell Wall. Notice that these bacteria are bounded by two membranes, the plasma membrane and the outer membrane of the cell wall.

→ **LPS** → Lipopolysaccharide

- Consists of three parts:

1- **lipid A**

2- **core polysaccharide**

3- **O side chain (O antigen)**

- Lipid A embedded in OM

- Core polysaccharide, O side chain extend out from the cell

- Importance of LPS:

- Contributes to negative charge on cell surface هو السبب في الشحنة السالبة للجرام (كثير مهمة) نيجتف
- Helps stabilize OM structure

- يساعد في جعل ال OM مستقر

- Attachment to surfaces and biofilm formation

- تساعد الخلية البكتيرية بأنها تضل مرتبطة بالسطح الي هي عليه وبساعدها بتشكيل ال Biofilms (رح يتم شرحهم لاحقا)

- Creates a permeability barrier

- يُعتبر كحاجز نفاذية

- Host defense protection (O antigen)

- ال O antigen يساهم في حماية البكتيريا المضرة من اليات دفاع المضيف تبعتها

- Acts as an endotoxin (lipid A)

- ال lipid A منقدر نعتبرو مادة سامة منسميها endotoxin

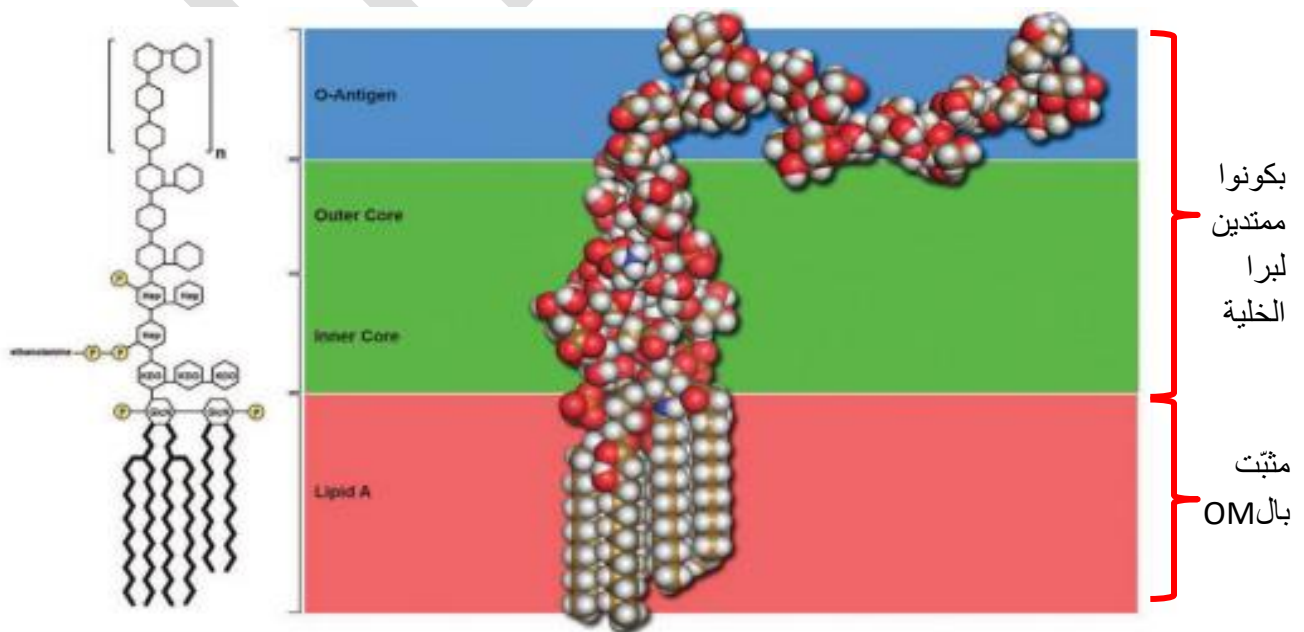


Figure 3.25 Lipopolysaccharide Structure. (a) A simplified diagram of LPS. Abbreviations: GlcN, glucosamine; Hep, heptulose; KDO, 2-keto-3-deoxyoctonate; P, phosphate. Lipid A is buried in the outer membrane. (b) Molecular model of a lipopolysaccharide. The lipid A and core polysaccharide are straight; the O side chain is bent at an angle in this model.

→ Gram-Negative Outer Membrane Permeability

- More permeable than plasma membrane due to presence of **porin** proteins and transporter proteins
- porin proteins form channels to let small molecules (600-700 daltons) pass

الـ OM نفاذيتو اكبر من النفاذية تبعت الـ plasma membrane
..ليه؟

لانو فيه بروتينات منسميها **Porins** بالاضافة لبروتينات ناقلة.
هدول الـ **porins** همة عبارة عن عن قنوات مليانة مي بتسمح لمركبات
جدا صغيرة انها تمرق من خلالها

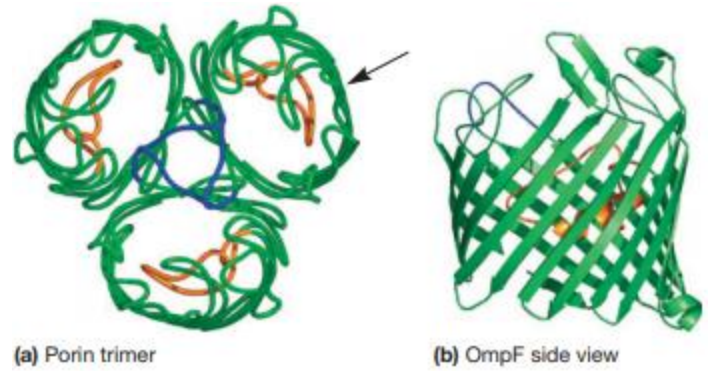


Figure 3.26 Porin Proteins. Two views of the OmpF porin of *E. coli*. (a) Structure of a trimeric porin observed when looking down at the outer surface of the outer membrane (i.e., top view). The center of each porin monomer is a water-filled channel. Each OmpF monomer can be divided into three loops: the green loop forms the channel, the blue loop interacts with other porin proteins to help form the trimer, and the orange loop narrows the channel. The arrow indicates the area of a porin molecule viewed from the side in panel (b). Side view of a porin monomer showing the β -barrel structure characteristic of porin proteins.

→ Mechanism of Gram Stain Reaction (الخطوات باختصار.. رح تاخذوها باللاب وتزهقوا منها)

1. Pass the loop over the bacterial colonies
2. Move the bacteria from the loop to the slide
3. Pass the slide over Bunsen burner for fixation بتثبت البكتيريا على السلايد من خلال الحرارة
4. Add crystal violet → (goes into the membrane)
5. Wash excess crystal violet with water
6. Add iodine (forms iodine-dye complex)
7. Wash excess iodine with tap water
8. Add alcohol for decolorizing → this closes the pores (they shrink) trapping the iodine-dye complex inside → in gram positive
9. Immediately wash with tap water
 - gram +ve : Violet
 - gram -ve: Colorless
10. Add sufranin
 - gram +ve: Violet
 - gram -ve : pink to red ($\text{⑥} > \text{⑤}$)

→ • Thinner peptidoglycan layer and larger pores of Gram-negative bacteria does not prevent loss of crystal violet

- alcohol may also remove/extract some lipids from outer layer of Gram-negative cell wall, making crystal violet dye removal easier

عاملين بخلوا خروج الـ crystal violet من الـ -ve اسهل من الـ +ve :

1. الـ peptidoglycan للـ -ve رفيع فيسمح للـ crystal violet انها تخرج بسهولة موزي تبع الـ -ve

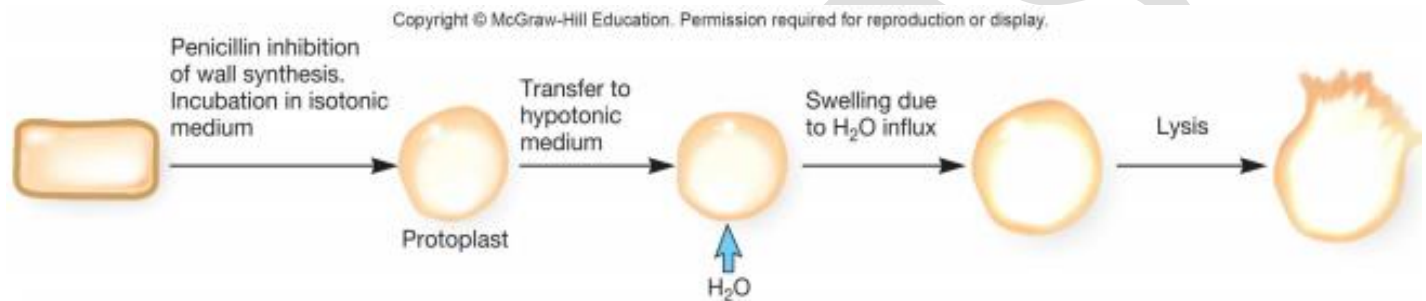
2. الكحول بذوب شوي من الـ lipids تبعت الـ OM فبصير اسهل للـ crystal violet انها تخرج

Osmotic Protection

• Hypotonic environments

- solute concentration outside cell less than inside cell
- water moves into cell and cell swells (bursts)
- cell wall protects from lysis
- Evidence for protection of cell wall:
 - Cells can be treated with:
 - Lysozyme → breaks bond between NAG and NAM
 - Penicillin → inhibits peptidoglycan synthesis

➡ If cells are treated with either of the above, they lyse in a hypotonic solution



• Hypertonic environments

- solute concentration outside cell is greater than inside
- water leaves cell (becomes shrunk)
- plasmolysis occurs

• Cells that Lose a Cell Wall May Survive in Isotonic Environments

- Isotonic → the concentration outside the cell equals the concentration inside

- Protoplasts (when the cell wall of gram +ve bacteria is treated with penicillin and destroyed..we call this cell a protoplast)
- Spheroplasts (when the cell wall of gram -ve bacteria is destroyed and the outer membrane remains we call this cell a Spheroplast)
- *Mycoplasma*
 - it is a bacteria that does not produce a cell wall
 - its plasma membrane is more resistant to osmotic pressure because it contains Cholesterol and it is the only type of bacteria that has it.

كيف تفرّق بين Hypotonic و Hypertonic (*٨-٨*)

احفظ الاولى وتحصيل حاصل حتكون عارف الثانية
Hypotonic : من حرف ال O تخيل اشي دائري بدو
ينفجر (الي هي الخلية بهاي الحالة)
بما انو بدو ينفجر معناته اخذ مي من برا كثير..والمي
بتتحرك من الاعلى تركيز للاقل تركيز وهيك بتعرف انو برا
الخلية هو الي فيه تركيز المي اكثر (يعني مادة ذائبة اقل)
وجوا فيه اقل (مادة ذائبة اكبر)
شو عكس الي حكيناها؟ انو يكون جوا الخلية فيه مي اكثر
وبرا اقل ف رح تتحرك المي من الداخل للخارج وهيك
الخلية رح يصير لها plasmolysis وهاي هي
hypertonic environments ال

3. Layers Outside the Cell wall

- They are on the outermost layer in the cell envelope
 - Gram +ve: on the peptidoglycan
 - Gram -ve: on the outer membrane
- Glycocalyx:
 - Any layer after the cell wall
 - Composed of polysaccharides
 - Important for the attachment of bacteria to each other and to host tissues or solid surfaces (e.g. biofilms in plants and animals)
 - They are:
 - i. Capsules
 - ii. Slime layers
 - iii. S layers

i. Capsules

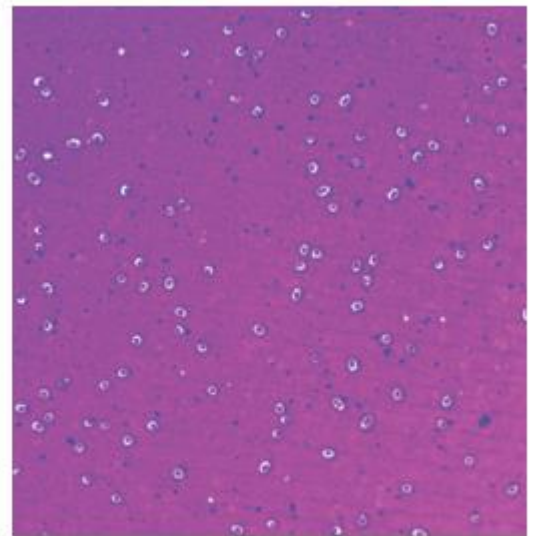
- Usually composed of polysaccharides
- Well organized and not easily removed from cell (opposite to slime layers)
- Visible in light microscope
- Protective advantages
 - resistant to phagocytosis(بلعمة)
 - protect from desiccation(الجفاف)
 - exclude viruses and detergents

ii. Slime layers

- Similar to capsules except that they are diffuse, unorganized and easily removed
- Slime may aid in motility

iii. S-layers

- Highly organized layers of protein or glycoprotein
- Have a very important property→ **Self-assembly** : they don't need any external information to gather its components
 - in Gram-negative bacteria S layer adheres to outer membrane
 - in Gram-positive bacteria, associated with peptidoglycan



K. pneumoniae

Figure 3.28 Bacterial Capsules. Capsule stain (a negative stain) of *Klebsiella pneumoniae*, bright-field light microscopy (×1,000).

- S Layer functions
 - protect from ion and pH fluctuations, osmotic stress, enzymes, and predation
 - maintains shape and rigidity
 - promotes adhesion to surfaces
 - protects from host defenses
 - potential use in nanotechnology
 - S layer spontaneously associates
 - Ex. *Deinococcus Radiodurans*: it has a high resistance to radiation

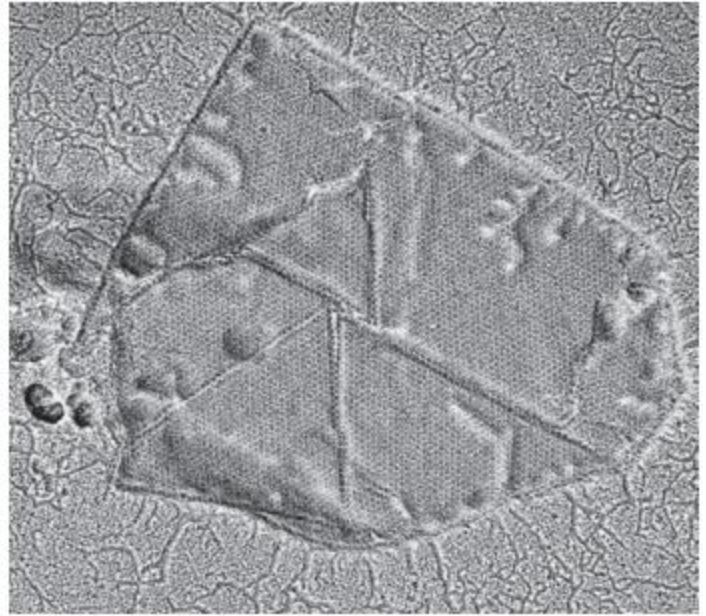


Figure 3.29 The S-Layer. An electron micrograph of the S-layer of the bacterium *Deinococcus radiodurans* after shadowing.

🌸 Bacterial cytoplasmic structures

📍 B. The Cytoplasm

- Bacterial Cytoplasmic Structures
 - I. Cytoskeleton
 - II. Intracytoplasmic membranes
 - III. Inclusions
 - IV. Ribosomes
 - V. Nucleoid and plasmids
- Cytoplasm : a material bounded by the plasma membrane
- Protoplast: is plasma membrane and everything within

I. The Bacterial Cytoskeleton

- Homologs of all 3 eukaryotic cytoskeletal elements have been identified in bacteria
من المعروف انو ال eukaryotes عندهم 3 عناصر اساسية لل cytoskeleton الي همة microtubules, intermediate filaments and microfilaments. وجدوا انو البكتيريا عندها عناصر كثير بتشبه هذول مع اختلافات بسيطة
- Functions are similar as in eukaryotes
حتى الوظائف تبعهم بتشبه الي موجودة بال eukaryotes
- We can consider them as roads and tracks inside the cell

Table 3.2 Some Bacterial Cytoskeletal Proteins

Type	Function	Comments
<i>Tubulin Homologues</i>		
FtsZ	Cell division	Widely observed in bacteria and archaea
BtubA/BtubB	Unknown	Observed only in <i>Prostheco bacter</i> spp.; thought to be encoded by eukaryotic tubulin genes obtained by horizontal gene transfer. In some species, Btub proteins assemble to form microtubule-like structures.
TubZ	Possibly plasmid segregation	Encoded by large plasmids observed in members of the genus <i>Bacillus</i>
<i>Actin Homologues</i>		
MamK	Positioning magnetosomes	Observed in magnetotactic species
MreB/Mbl	Helps determine cell shape; may be involved in chromosome segregation, localizing proteins, motility, and establishing cell polarity	Most rod-shaped bacteria
ParM	Plasmid segregation	Plasmid encoded
<i>Intermediate Filament Homologues</i>		
CreS (crescentin)	Induces curvature in curved rods	<i>Caulobacter crescentus</i>
<i>Unique Bacterial Cytoskeletal Proteins</i>		
MinD	Prevents polymerization of FtsZ at cell poles	Many rod-shaped bacteria
ParA	Segregates chromosomes and plasmids, helps localize chemotaxis proteins and type IV pili to one pole of certain rod-shaped bacteria	Observed in many species, including <i>Vibrio cholerae</i> , <i>C. crescentus</i> , and <i>Thermus thermophilus</i>

- Three important structures of the bacterial cytoskeleton

1- FtsZ:

- found in many bacteria
- very important during cell division. It forms a ring at the center of a dividing cell and is required for the formation of the **septum** that will separate the daughter cells → where there's a binary fission

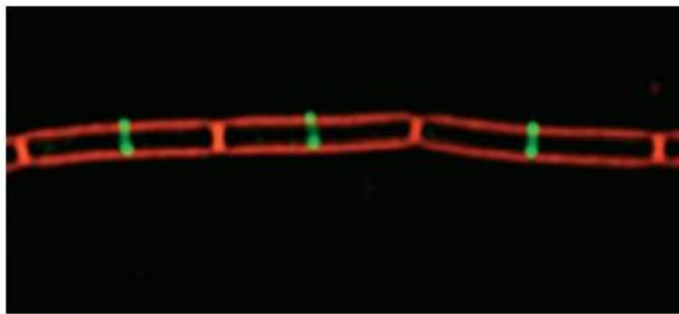
2- MreB:

- Found in Rods (البكتيريا التي على شكل قضبان)
- Maintains the shape of rods by positioning peptidoglycan synthesis machinery

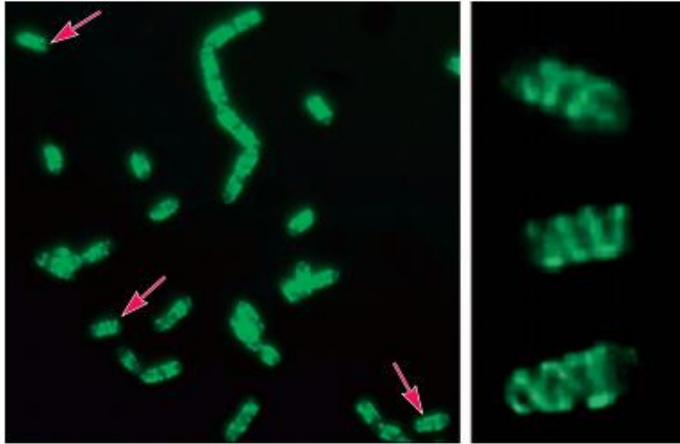
يحافظ على الشكل من خلال التحكم بآلية تصنيع الـ peptidoglycan

3- CreS

- Rare, only found in vibrio cells (التي على شكل فاصلة)
- Important for maintaining its curve

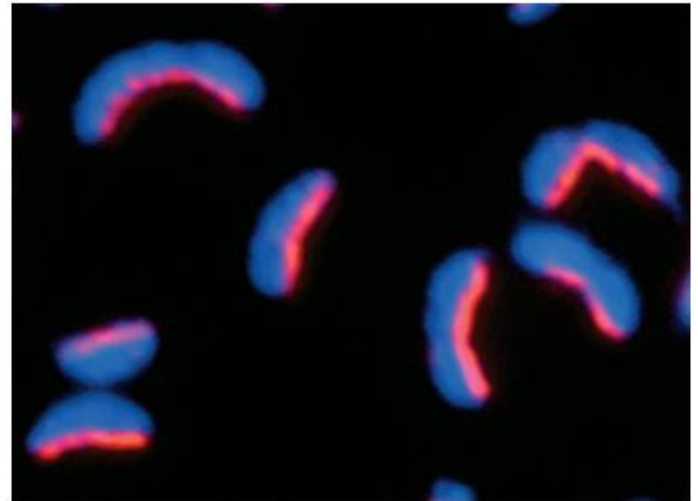


(a) FtsZ



(b) Mbl

(c) Mbl



(d) Crescentin

Figure 3.30 The Bacterial Cytoskeleton. (a) FtsZ protein in a chain of *Bacillus subtilis* cells; FtsZ-green fluorescent (GFP) fusion protein viewed by fluorescence microscopy. (b) The MreB-like cytoskeletal protein (Mbl) of *B. subtilis*; Mbl-GFP in live cells was examined by fluorescence microscopy. With this method, Mbl seems to form helices (arrows). More sensitive methods show that Mbl forms patches that move perpendicular to the long axis of the cell. (c) Three of the cells from (b) are shown at a higher magnification. (d) CreS (crescentin), in red, of *Caulobacter crescentus*. The DNA in the cells was stained blue with DAPI.

II. Intracytoplasmic membranes

→ Plasma membrane infoldings

- Why would bacteria have multiple membranes?

More membranes → More surface area

- observed in many photosynthetic bacteria
- observed in many bacteria with high respiratory activity

Intra:
inside and
within
Inter:
Between

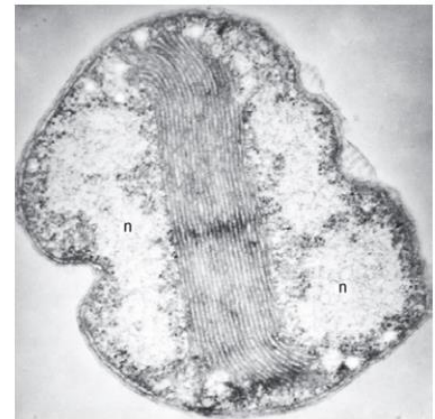


Figure 3.31 Internal Bacterial Membranes. The nitrifying bacterium *Nitrocystis oceanus* has parallel membranes traversing the whole cell. Note the nucleoid (n) with fibrillar structure.

III. Inclusions

→ Granules of organic or inorganic material that are stockpiled (تخزينها) by the cell for future use

(تذكروا انو البكتيريا بتعيش ب بيئات فقيرة بالغذاء ليهيك لما يكون الغذاء متوفر بتاخده وبتخزنه عندها بتراكيب اسمها
(inclusions)

- Some are enclosed by a single-layered membrane
 - membranes vary in composition
 - some made of proteins; others contain lipids
- may be referred to as microcompartments
- Types of Inclusions:

(a) Storage inclusions → Storage of nutrients, metabolic end products, energy, building blocks.

- Glycogen storage (glycogen is a glucose polymer)
- Carbon storage: stored in the form of Poly- β -Hydroxybutyrate (PHB)
- Phosphate : stored as Polyphosphate called **Volutin**
- Amino acids: they are the source of nitrogen. They are stored as cyanophycin granules → usually these amino acids are Aspartic acid and Arginine

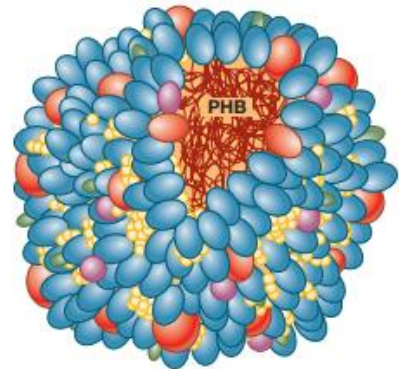


Figure 3.33 PHB Inclusions in Bacteria. PHB is a polymer of β -hydroxybutyrate molecules joined by ester bonds between the carboxyl and hydroxyl groups of adjacent molecules. PHB inclusions are around 0.2 to 0.7 μm in diameter. PHB is enclosed by a shell composed of several different proteins, including the PHB-synthesizing enzyme (red sphere) and the PHB-degrading enzyme (green sphere). Yellow spheres represent the phospholipids that are also found in the shell. Note that these phospholipids do not form a phospholipid bilayer.



Figure 3.34 Sulfur Globules. *Chromatium vinosum*, a purple sulfur bacterium, with intracellular sulfur globules, bright-field microscopy ($\times 2,000$).

(b) Enzyme storage inclusions (Microcompartments)

→ They are NOT bound by membranes but compartments for specific functions

- Carboxysomes → found in CO_2 fixing bacteria
→ Store the enzyme **Rubisco** for carbon fixation

Carbon fixation and Photosynthesis are the same

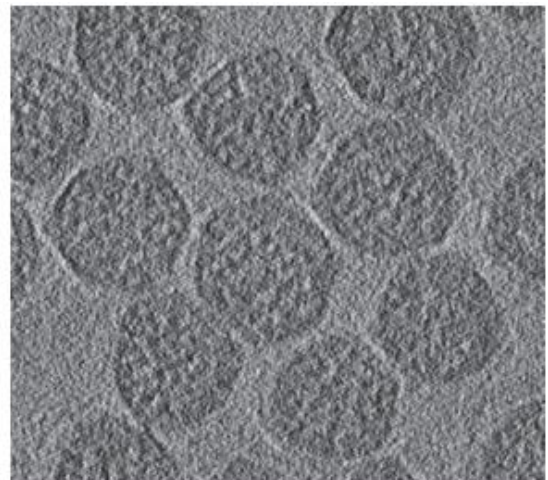


Figure 3.35 Carboxysomes. Carboxysomes in the bacterium *Halothiobacillus neapolitanus*. This is one image of a tilt series taken for electron cryotomography. Each carboxysome is approximately 100 nm in diameter.

(c) Gas vacuoles

- Found in aquatic, photosynthetic bacteria and archaea
- Provide buoyancy (طفو) in gas vesicles: Photosynthetic bacteria need to be at the surface to absorb light and to exchange gases
- Gas vacuoles look like cigarettes



Figure 3.36 Gas Vacuoles Are Clusters of Gas Vesicles. A freeze-fracture preparation of *Anabaena flosaquae* ($\times 89,000$) showing gas vesicles and gas vacuoles. Both longitudinal and cross-sectional views of gas vesicles are indicated by arrows.

(d) Magnetosomes

- Found in aquatic bacteria
- Magnetite particles that orient bacteria toward North & South poles (☉_☉) !
- In them is a cytoskeletal protein that is only found in them called MamK, it helps in forming magnetosome chain

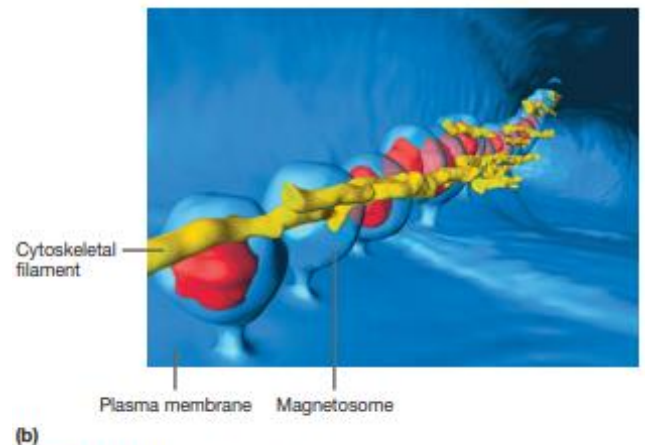
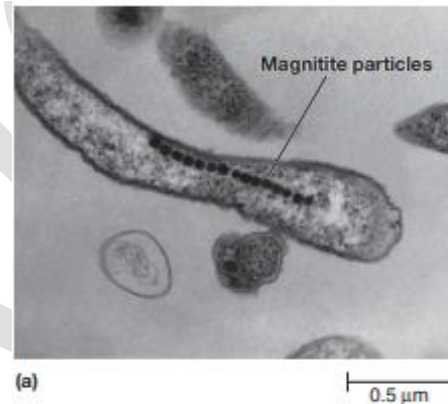


Figure 3.37 Magnetosomes. (a) Transmission electron micrograph of the magnetotactic bacterium *Magnetospirillum magnetotacticum*. (b) Three-dimensional reconstruction of the magnetosomes of *M. magnetitum* (electron cryotomograph).

IV. Ribosomes

- Protein factories
- In eukaryotes and prokaryotes, ribosomes have the same Function but different Structure
- Ribosomes consist of TWO subunits:
 - * A large subunit 50 S → هدول ثابتين
 - * A small subunit 30 S → prokaryotes لل
- Prokaryotic ribosome : 70 S
- Eukaryotic ribosome: 80 S

ما منجمع ال small وال large subunits عشان نطلع ال prokaryotic وال eukaryotic ribosomes

- Further precipitation for the 30s subunit → 16S rRNA
- Further precipitation for the 50s subunit → 23S and 5S rRNA

S: Svedberg unit → for precipitation

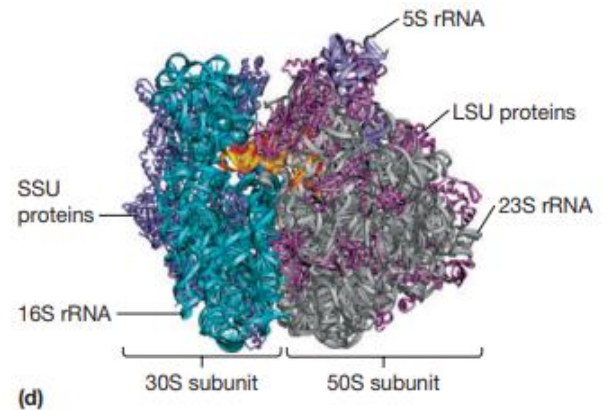
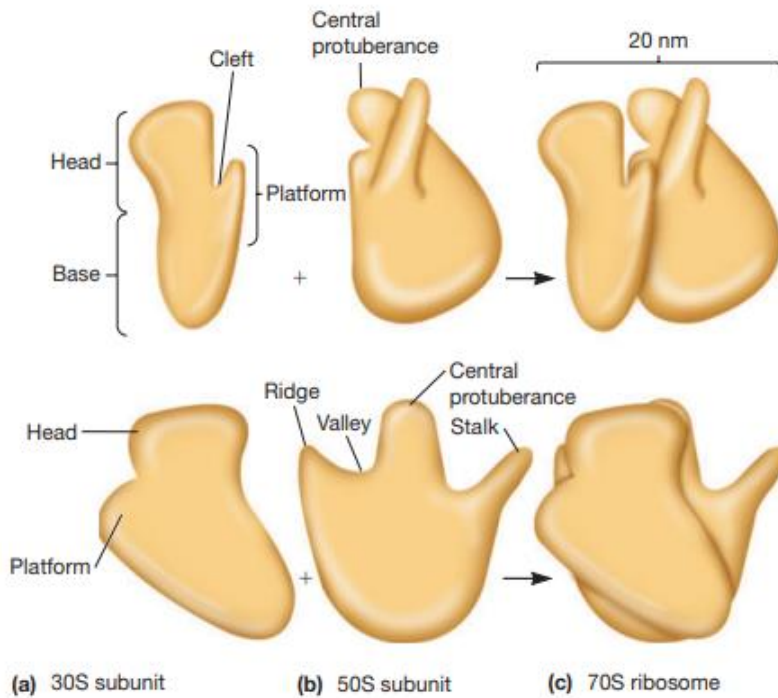


Figure 3.38 Bacterial Ribosomes. (a–c) Schematic representation of the two subunits and the complete 70S ribosome of *E. coli*. (d) Molecular structure of the 70S ribosome of *Thermus thermophilus*. The 50S subunit (LSU) includes 23S rRNA (gray) and 5S rRNA (lavender), while 16S rRNA (turquoise) is found in the 30S subunit (SSU). A molecule of tRNA (gold) is shown in the A site. To generate this ribbon diagram, crystals of purified bacterial ribosomes were prepared and exposed to X rays, and the resulting diffraction pattern analyzed.

V. Nucleoid and Plasmids

- The Nucleoid:

→ Usually not membrane bound (few exceptions)

→ Location of the chromosome and associated proteins

(The main difference between Eukaryotes and Prokaryotes is that eukaryotes have their genetic material enclosed by a membrane whereas in prokaryotes it's not, and the region where the genetic material exists is called The Nucleoid.)

→ the genetic material is usually 1 closed circular, double-stranded DNA molecule

ولكن هو فعليا لما تطلع عليه بالميكروسكوب ما رح تشوفه دائرة لأنو كثير منطوي على بعضو. القصد بأنو دائري يعني النهاية من الجهتين مسكرة على بعضها

• Supercoiling and nucleoid proteins

(different from histones) aid in folding

الي يساعد على هاد الطوي هي بروتينات موجودة بهاي المنطقة (مو الهيستون الي منلاقه بال eukaryotes)

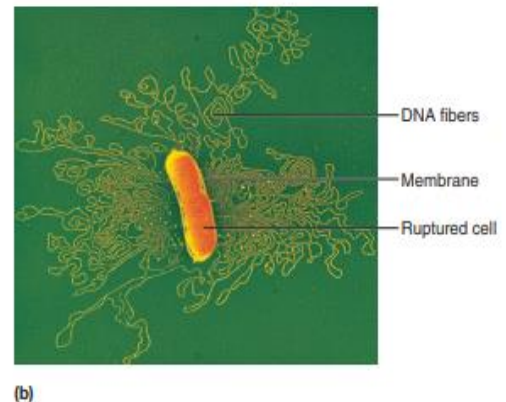
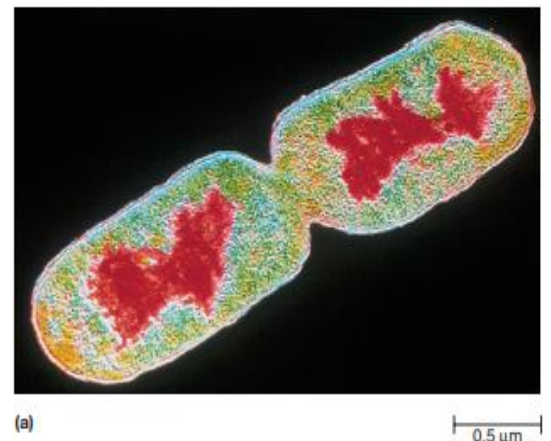


Figure 3.39 *E. coli* Nucleoids and Chromosomes. Bacterial chromosomes are located in the nucleoid, an area in the cytoplasm. (a) A color-enhanced transmission electron micrograph of a thin section of a dividing *E. coli* cell. The red areas are the nucleoids present in the two daughter cells. (b) Chromosome released from a gently lysed *E. coli* cell. Note how tightly packaged the DNA must be inside the cell.

- Plasmids

- It is an Extrachromosomal DNA (يعني هو اشي والكروموسوم اشي مختلف تماما)
 - found in bacteria, archaea, some fungi
 - usually small, closed circular DNA molecules
- they carry genes, but these genes are not important in Growth or Production
- Exist and replicate independently of chromosome
- Two important properties of plasmids:
 - **Episomes**: Plasmids can integrate with the chromosome and be part of it
 - **Curing**: the loss of the plasmid during cell division
- Plasmids have many functions:
 - 1) Conjugative plasmids: Transfer DNA from one cell to another through an external structure called Pilus. The transfer DNA may go to another cell that is totally different from the first one (a different species)
→ *E.coli, Salmonella, Citrobacter*
 - 2) R plasmids: for Resistancy
→ *Pseudomans* & many gram -ve
 - 3) Col plasmids: produce **Bacteriocins** → toxins that destroy other nearby bacterial populations (a fight for resources)
 - 4) Virulence plasmids: factors that the bacteria use to colonies the host tissue and to evade the immune system
→ *Agrobacterium*: induces cancer in plants
 - 5) Metabolic plasmids: have genes that allow the bacteria to degrade compounds found in the environment that are difficult to be degraded otherwise.
→ *Pseudomonas*

Table 3.3 Major Types of Bacterial Plasmids

Type	Function	Example	Size (kbp)	Hosts	Phenotypic Features ¹
Conjugative Plasmids²	Transfer of DNA from one cell to another	F factor	95–100	<i>E. coli, Salmonella, Citrobacter</i>	Sex pilus, conjugation
R Plasmids	Carry antibiotic-resistance genes	RP4	54	<i>Pseudomonas</i> and many other Gram-negative bacteria	Sex pilus, conjugation, resistance to Amp, Km, Nm, Tet
Col Plasmids	Produce bacteriocins, substances that destroy closely related species	ColE1	9	<i>E. coli</i>	Colicin E1 production
Virulence Plasmids	Carry virulence genes	Ti	200	<i>Agrobacterium tumefaciens</i>	Tumor induction in plants
Metabolic Plasmids	Carry genes for enzymes	CAM	230	<i>Pseudomonas</i>	Camphor degradation

¹ Abbreviations used for resistance to antibiotics: Amp, ampicillin; Gm, gentamycin; Km, kanamycin; Nm, neomycin; Tet, tetracycline.

² Many R plasmids, metabolic plasmids, and others are also conjugative.

External structures and motility

- External structures: They extend beyond the cell envelope in bacteria
→Function in:
Protection, Attachment to surfaces, Horizontal gene transfer, Cell movement
- We have:
 1. Pili and Fimbriae
 2. Flagella

1. Pili and Fimbriae

- Fimbriae (s., fimbria);
 - Short, thin, hairlike, protein appendages (up to 1,000/cell)
 - can mediate attachment to surfaces, motility, DNA uptake
- Sex Pili (s., pilus)
 - longer, thicker, less numerous (1-10/cell)
 - genes for formation on plasmids
 - required for conjugation

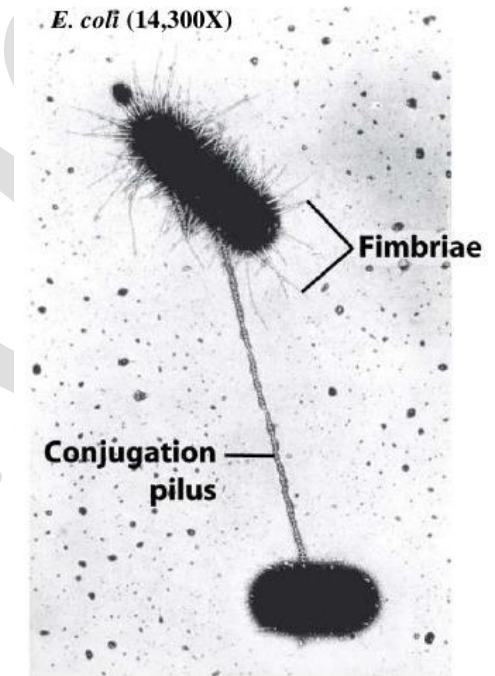


Figure 4-16 Microbiology, 6/e
© 2005 John Wiley & Sons



Figure 3.40 Flagella and Fimbriae. The long flagella and numerous shorter fimbriae are evident in this SEM of the bacterium *Proteus vulgaris*.

Difference between Pili and fimbriae:

- Fimbriae are shorter than Pili.
- The diameter of Pili is higher than that of fimbriae.
- A cell can have 1 to 10 pili and about 200 to 300 fimbriae.
- Fimbriae are specialized for attachment of bacteria cell to a host, whereas pili are responsible for bacterial conjugation.

2. Flagella

- Threadlike, locomotor appendages (tail) extending outward from plasma membrane and cell wall
- Thin, rigid protein structures that cannot be observed with bright-field microscope unless specially stained

- Functions:

- motility and swarming behavior*
- attachment to surfaces
- may be virulence factors (هاي عوامل ممكن تزيد من عدائية البكتيريا)

**only seen in the lab when its moving on a surface that is not liquid nor solid*

→ agar

- Patterns of flagellation: (how many and where they're positioned)

- Monotrichous → one flagellum
- Polar flagellum → flagellum at end of cell
- Amphitrichous → one flagellum at each end of cell
- Lophotrichous → cluster of flagella at one or both ends
- Peritrichous → spread over entire surface of cell

-Trichous → hair

Amphi- → Two

Lopho- → Cluster

Peri- → Around

- Ultrastructure composed of three parts

- A **filament** that extends from cell surface to the tip
 - it is a hollow, rigid cylinder of flagellin protein
- a **Hook** that links filament to basal body
- a **Basal body** which is a series of rings that drive flagellar motor

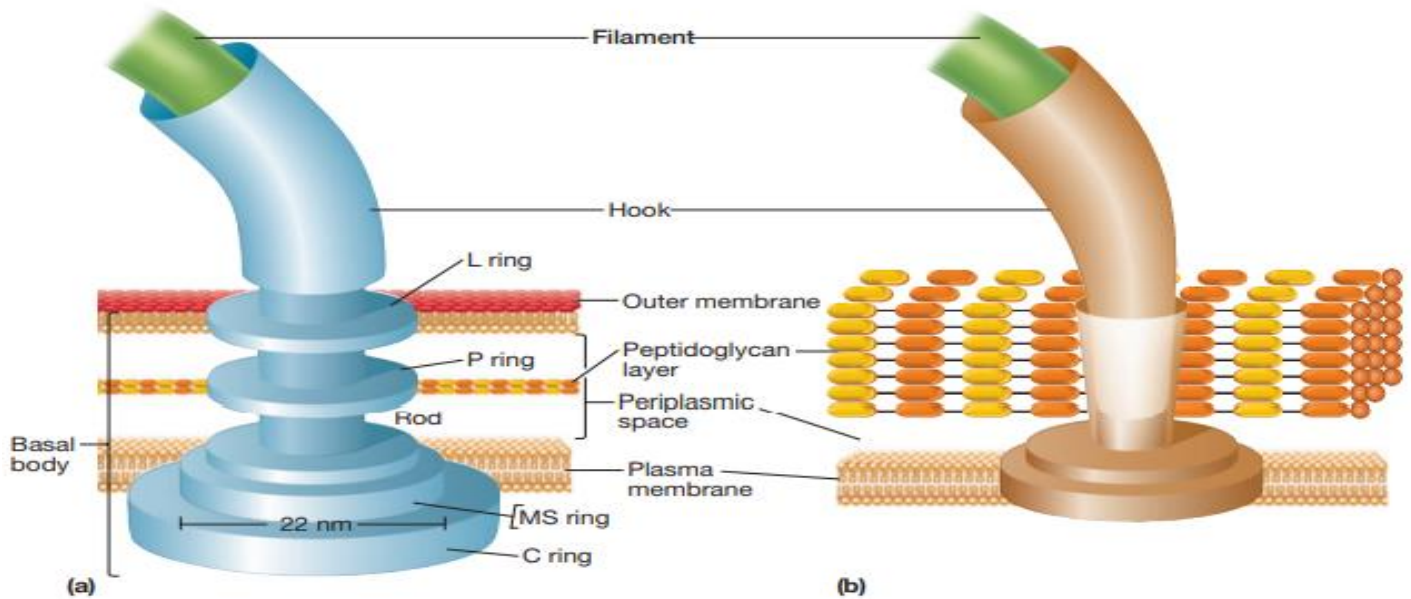
تركيب ال basal body يختلف بين ال gram positive وال gram negative:

(a) In Gram Negative → 4 Rings

- 1- **L** ring: refers to Lipopolysaccharide; exists in the outer membrane
- 2- **P** ring: embedded in Peptidoglycan layer
- 3- **S** ring: in the Periplasmic space
- 4- **M** ring: in plasma membrane

(b) In Gram Positive → 2 rings

- 1- **P** ring: in peptidoglycan
- 2- **M** ring: in plasma membrane



- ❖ *Helicobacter pilory* can live in the stomach but not in the stomach juice where the PH=2. It lives on the stomach linings where the PH=7 since it has a flagellum that helps it to swim there and a urease enzyme that breaks down urea inside the stomach juice transforming it to ammonia (NH₃). Ammonia reacts with HCl forming micro-neutral PH at the surface of the bacteria; this happens only when it's swimming to reach the stomach linings

- Flagellar Synthesis

- It is a complex process involving many genes/gene products

- Flagellin proteins make Filaments

→ Flagellin proteins have two faiths

- (1) In bacteria, new flagellin molecules are transported all the way through the hollow filament and are added at the tip using Type III-like secretion system

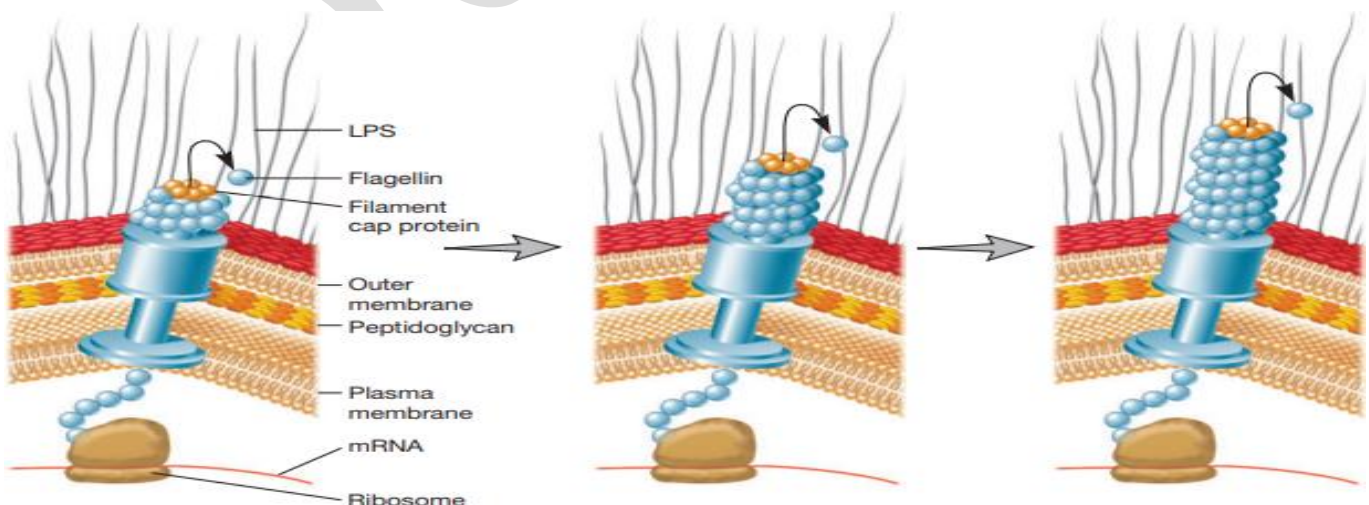


Figure 3.43 Flagellar Filaments Grow at the Tip. Flagellin subunits travel through the flagellar core and attach to the growing tip. Their attachment is directed by the filament cap protein.

(2) In Archea, Flagellin molecules are added at the base

- Filament subunits **self-assemble** with help of filament cap at tip, not base

• Motility

i. Flagellar movement:

- Flagellum rotates like a propeller(مروحة)
→ It is a very rapid rotation → up to 1100 revolutions/sec

It can either be:

- Counter Clock Wise (CCW) rotation → causes Forward motion(run) → away from toxins
→ toward food
- Clock Wise (CW) rotation → disrupts run causing cell to stop and tumble
→ (when there's noo specific nutrient to go for or a toxin to run away from it tumbles)

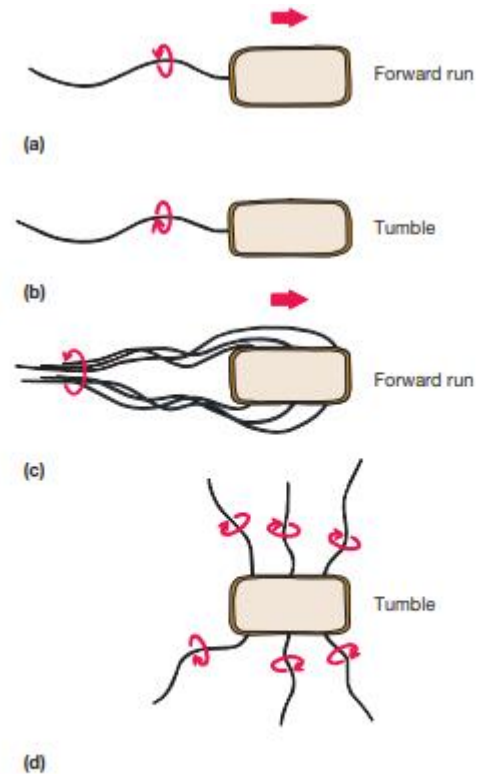


Figure 3.44 The Direction of Flagellar Rotation Often Determines the Way a Bacterium Moves. Parts (a) and (b) describe the motion of many monotrichous, polar bacteria. Parts (c) and (d) illustrate the movements of many peritrichous organisms.

Mechanism of Flagellar Movement
is NOT required-Dr.Sreen

ii. Swarming

- Occurs on moist or semisolid surfaces in the lab as a type of group behavior by bacteria
- Most swimmers have peritrichous flagella
- Production of molecules that aid movement is typical

iii. Spirochete Motility

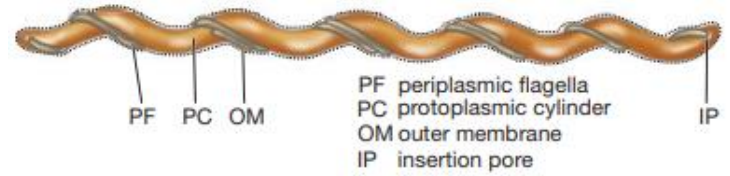
- Multiple flagella form **axial fibril** which winds around the cell
- Flagella remain in periplasmic space inside outer sheath(see figure 3.47)
- Corkscrew shape exhibits flexing and spinning movements



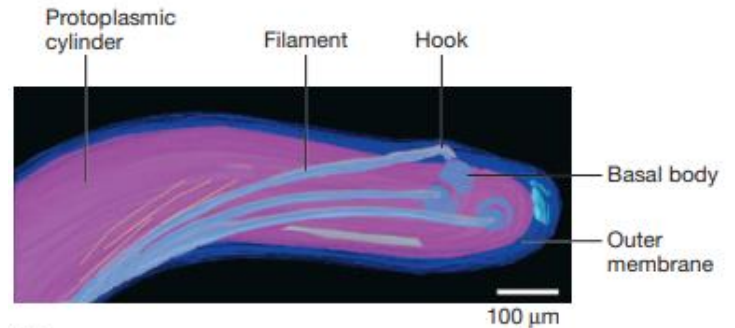
Figure 3.46 Swarming Bacteria Often Produce Distinctive Patterns on a Solid Growth Medium. These bacterial cells swarmed out from the center of the plate and produced a branching pattern called dendrites.

- Helps the bacteria in its crawling motility and it has to be attached to a solid surface

التركيب مهم والاسماء مهمة



(a)



(b)

Figure 3.47 Spirochete Flagella. (a) Numerous flagella arise from each end of the spirochete. These intertwine and wind around the cell, usually overlapping in the middle. (b) Electron cryotomographic image of the spirochete *Treponema denticola* showing three flagella arising from the tip of the cell

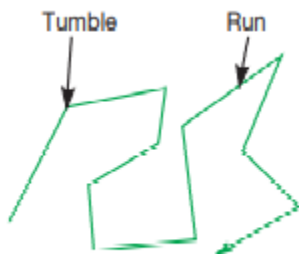
- iv. Twitching and Gliding Motility
 - May involve Type IV pili and slime
 - Twitching→irregular motions
 - pili at ends of cell
 - short, intermittent, jerky motions
 - cells are in contact with each other and surface
 - Gliding→smooth movements

v. Chemotaxis*

- Communication occur within populations by signals and those signals are chemical signals; those signals may come from other cells or from the environment
- Chemotaxis is movement toward a chemical attractant or away from a chemical repellent
- Changing concentrations of chemical attractants and chemical repellents bind chemoreceptors of chemosensing system
- In presence of attractant/repellant, tumbling frequency is reduced; runs toward/away from compound are longer

بغيا chemical gradient ، الـ *E.coli* يتمشي بشكل عشوائي وبتضل تبدل بين الـ running والـ tumbling . خلال الـ running بتكون البكتيريا يتمشي بخط مستقيم او منحني شوي وبعدين بتوقف وبتعمل tumbling . هاد الـ tumbling بعيد توجيه البكتيريا وبتصير مواجهة لجهة مختلفة ليهيك لما ترجع تعمل running بتكون عم يتمشي باتجاه مختلف. (figure 3.48a)

لما تكون البكتيريا معرضة لـ attractant الـ tumbling بصير اقل ويكون الـ running لمسافات اطول (لما تكون يتمشي باتجاه هاد الـ attractant) (figure 3.48b). ونفس الشي بس بعيدا عن المادة لما يكون في repellent ... يعني الـ running يكون لمسافات اطول والـ tumbling اقل بس باتجاه معاكس.



(a)



(b)

Figure 3.48 Tumbles and Runs Are Used to Direct the Movement of Many Bacteria Toward a Chemical Attractant. (a) Random movement of a bacterium in the absence of a concentration gradient. Tumbling frequency is fairly constant. (b) Movement in an attractant gradient. Tumbling frequency is reduced when the bacterium is moving up the gradient. Therefore runs in the direction of increasing attractant are longer.

- bacteria exhibit taxis to a variety of stimuli, including light and oxygen

*Chemo Taxis→Chemo: Chemical
→Taxis: Attract

- Behavior of bacterium altered by temporal concentration of a chemical

🌸 Bacterial endospores

Bacterial Endospores Are a Survival Strategy

- Bacterial endospores are Complex, **dormant** (ساکن) structure formed by some bacteria
- They are formed inside the cell
- They are positioned in various locations within the
→its position is so important in bacterial classification
- They are highly resistant to numerous environmental conditions:
 - Heat
 - Radiation
 - Chemicals
 - Desiccation
- Only gram positive bacteria can form endospores (but not all of them)
- Bacteria can stay dormant for 100,000 years; when the environment conditions are more suitable, it continues its growing
- The process of endospore formation is called **Sporulation**
 - Occurs in hours (up to 10 hours)
 - Normally commences when growth ceases
 - because of lack of nutrients
 - Complex multistage process
 - They are made of calcium (complexed with dipicolinic acid)

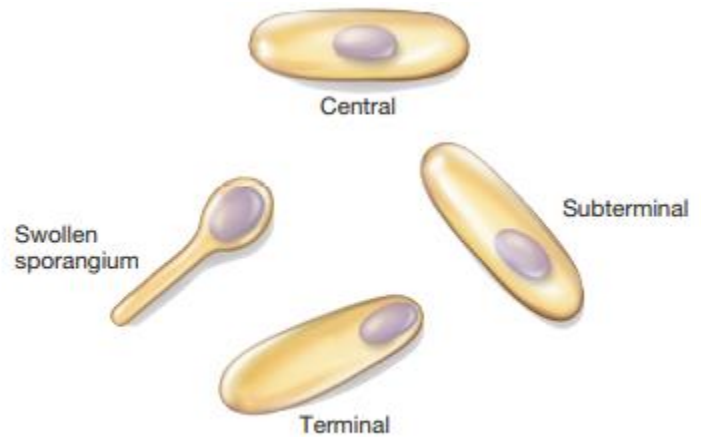
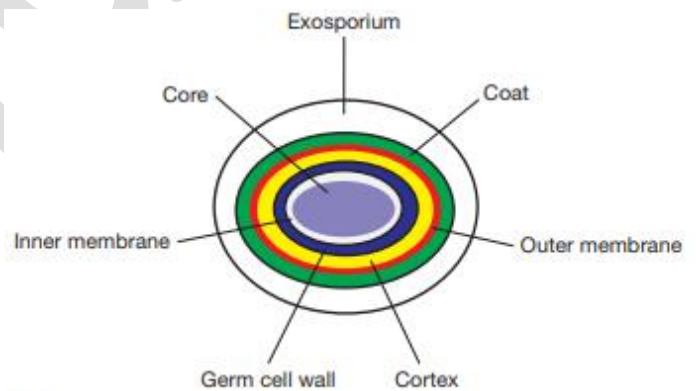
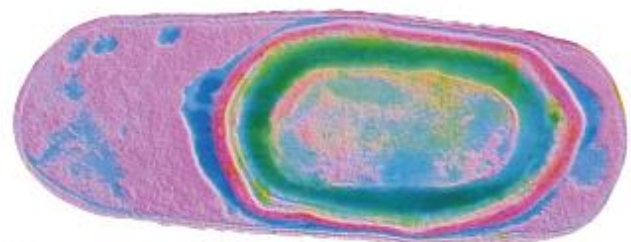


Figure 3.50 Examples of Endospore Location and Size.



(a)



(b)

Figure 3.49 Bacterial Endospores. (a) A schematic showing the various layers of an endospore. They are not drawn to scale. (b) A colorized cross section of a *Bacillus subtilis* cell undergoing sporulation. The oval in the center is an endospore that is almost mature; when it reaches maturity, the mother cell will lyse to release it.

- DNA is protected by small acid soluble DNA binding proteins (SASP)
- Have a dehydrated core (which means that there's not much water in the core)
- spore coat and exosporium protect
- Metabolically inactive
- Formation of vegetative cells:

A- Activation

- Activation is a process that prepares spores for germination
- often results from treatments like heating

(عشان ترجع الخلية لوضعها الطبيعي لازم تتعرض للحرارة...قد ما وفرناها غذاء، مستحيل ترجع الا لما تتعرض للحرارة)

B- Germination

- Germination: the breaking of the spore's dormant state
- environmental nutrients are detected
- spore swelling and rupture of absorption of spore coat
- increased metabolic activity

C- Outgrowth : emergence of vegetative cell

The End

(~▽~)~ ~(^▽^)~